GENERAL SCIENCE GRADE 10: MAGNETISM AND ELECTROMAGNETIC INDUCTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 3.0: Natural Physical Science
Sub-Strand	Sub-Strand 3.4: Magnetism and Electromagnetic Induction
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Magnetisation and demagnetisation of soft iron (electrical, induction, hammering and stroking methods) – Demagnetisation methods (electrical, hammering and heating) – Magnetic field patterns around a magnet – Induced electromotive force (e.m.f) in electromagnetism – Electromagnetic induction experiment (U-shaped magnet, galvanometer and straight conductor) – Factors affecting the magnitude of induced electromotive force (qualitative treatment) – Applications of electromagnetic induction in day-to-day life – Project: Design and make a simple electric bell using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the methods of magnetisation and demagnetisation in soft iron, b) describe the magnetic field patterns around a magnet, c) describe induced electromotive force in electromagnetism, d) perform an experiment on electromagnetic induction, e) explain factors affecting the magnitude of induced electromotive force (qualitative treatment), f) appreciate the applications of electromagnetic induction in day-to-day life.
Core Competencies	– Creativity and Imagination: as the learner designs and makes a simple electric bell using locally available materials such as scrap wire, a battery, a nail and metal strips sourced from Nairobi's Gikomba or Kamukunji market areas. – Learning to Learn: as the learner carries out experiments of magnetising a magnetic material and demagnetising a magnet, reflecting on results and adjusting experimental technique accordingly. – Critical Thinking and Problem Solving: as the learner investigates and compares the effectiveness of different magnetisation methods and selects the most suitable one based on evidence gathered. – Digital Literacy: as the learner uses available digital and print media to research on the factors that affect the magnitude of induced electromotive force and applications of electromagnetic induction.
Core Values	– Unity: as the learner carries out assigned roles when performing experiments to demonstrate electromagnetic induction, working collaboratively with group members. – Responsibility: as the learner takes care of the apparatus (U-shaped magnet, galvanometer and straight conductors) while

	– carrying out experiments on electromagnetic induction. – Integrity: as the learner honestly records and reports observations from magnetisation, demagnetisation and electromagnetic induction experiments without falsifying data. – Respect: as the learner appreciates and considers the contributions of peers during group discussions on the applications of electromagnetic induction in day-to-day life.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner formulates investigable questions about why a galvanometer deflects when a conductor moves through a magnetic field and what variables influence the size of the deflection. – Planning and Carrying Out Investigations: as the learner designs and conducts controlled experiments on magnetisation, demagnetisation and electromagnetic induction using a U-shaped magnet, galvanometer and straight conductor. – Analysing and Interpreting Data: as the learner records galvanometer deflection readings under different experimental conditions (conductor stationary, moving parallel, moving vertically, moving at an angle) and draws evidence-based conclusions. – Constructing Explanations: as the learner explains the induced e.m.f concept using experimental evidence and connects it to real-life applications such as generators and electric bells. – Obtaining, Evaluating and Communicating Information: as the learner researches factors affecting induced e.m.f using digital and print media and shares findings with peers in plenary discussions.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Self-Esteem): as the learner explains the methods of magnetisation and demagnetisation with confidence, demonstrating growing scientific self-efficacy. – Safety and Security: as the learner observes laboratory safety precautions when handling apparatus during electromagnetic induction experiments, including safe use of electrical connections. – Financial Literacy: as the learner uses locally available and low-cost materials (nails, scrap wire, dry cells) to construct a simple electric bell, appreciating cost-effective innovation. – Socio-Economic and Environmental Issues: as the learner explores how electromagnetic induction underpins electricity generation in Kenya, including hydroelectric power at stations such as those on the Tana River and geothermal plants in Olkaria, Naivasha.
Career Connections	– Electrical Engineer: electromagnetic induction is the foundational principle behind generator and transformer design used in Kenya Power's national grid infrastructure. – Renewable Energy Technician: wind and hydroelectric turbines at sites such as Lake Turkana Wind Power and Olkaria Geothermal Plant operate on electromagnetic induction principles covered in this sub-strand. – Telecommunications Technician: understanding magnetic fields and induced e.m.f supports work with antennas, signal transformers and communication infrastructure across Kenya. – Mechatronics and Automation Technician: electric motors and solenoid-based actuators used in Kenya's growing manufacturing and agri-processing industries rely on electromagnetism concepts. – Biomedical Equipment Technician: MRI machines and other hospital diagnostic equipment used in facilities like Kenyatta National Hospital involve magnetic field principles explored in this sub-strand.
Focus for Lessons	This unit investigates the principles of magnetism and electromagnetic induction through hands-on experimentation, structured inquiry and real-world application. Learners explore how magnetic fields are created and mapped, how motion within a magnetic field induces an electromotive force, and what variables control the magnitude of that induced e.m.f. The unit is anchored in Kenya's lived energy landscape — from hand-cranked torches used by farmers in rural Rift Valley to the massive turbines at Olkaria Geothermal Plant — helping learners connect abstract electromagnetic theory to tangible national development challenges.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya?

	KICD KEY INQUIRY QUESTION: 1. How is the study of electromagnetic induction important in our daily life?
Anchoring Phenomenon	A form one student in a boarding school in Kericho notices that when the school's generator room door is open during a power outage, she can see large copper coils rotating inside a metal housing connected to the national grid line. No battery or chemical reaction is visible, yet the coils produce enough electricity to light the entire dormitory. She is puzzled: how can simply spinning a coil of wire near a magnet produce electricity powerful enough to run lights, water pumps and kitchen equipment? She tries to replicate this at home using a bicycle dynamo she removes from an old bike — when she spins the wheel, a small LED lights up, but only when the wheel is moving. When she holds the dynamo still, the light goes out immediately. She cannot explain why movement is essential, why faster spinning makes the light brighter, or why the same principle that powers her bicycle light is also responsible for Kenya's national electricity supply.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Magnetisation Methods: Learners observe the bicycle dynamo and hand-cranked torch demonstrations, record initial puzzling observations, and make written predictions about the source of electricity. They then carry out experiments on methods of magnetisation of soft iron (stroking, electrical and induction methods), compare results, and begin the Driving Question Board (DQB) by posting "What do we notice?" and "What do we wonder?" questions. Lesson 2 — Demagnetisation & Properties of Magnets: Building on Lesson 1, learners carry out demagnetisation experiments (heating over a jiko, hammering on a hard surface, and electrical demagnetisation using alternating current). They compare how each method destroys magnetism and connect the Kericho student's observation that the generator coils must keep moving — stillness produces nothing — to the loss of ordered magnetic domains. Lesson 3 — Magnetic Field Patterns: Learners use iron filings and plotting compasses to map and draw field lines around bar magnets and U-shaped magnets. They observe field concentration at poles, field direction conventions, and the uniform field between the poles of a U-shaped magnet. They annotate field diagrams and update the DQB: "Where is the field strongest — and why does that matter for a generator?" Lesson 4 — Introducing Induced E.M.F (Qualitative): Learners are introduced to the concept of induced electromotive force. Using a U-shaped magnet, a sensitive galvanometer and a straight copper conductor, they carry out the KICD-prescribed electromagnetic induction experiment: observing deflection when the conductor is (a) stationary between poles, (b) moved parallel to field, (c) moved vertically upward, (d) moved at an angle to the field. Learners record deflection direction and magnitude for each condition and construct an initial explanation for the anchoring phenomenon. Lesson 5 — Factors Affecting Magnitude of Induced E.M.F: Learners design and conduct guided investigations varying (a) speed of conductor movement, (b) strength of the magnet, (c) number of turns of wire, and (d) angle of movement through the field. Results are tabulated and graphed. Learners connect findings back to the dynamo torch observation: faster cranking → larger induced e.m.f → brighter LED. The DQB is updated with evidence-based answers replacing earlier predictions. Lesson 6 — Applications of Electromagnetic Induction in Kenya: Learners use digital devices and print media to research real-life applications — AC generators, transformers, the Olkaria Geothermal Plant, Lake Turkana Wind Farm turbines, electric bells and bicycle dynamos. Groups prepare short presentations linking each application to the experimental principles from Lessons 4 and 5. Learners discuss how Kenya Power's national grid depends entirely on electromagnetic induction.

	Lesson 7 — Project: Design and Make a Simple Electric Bell: Learners design and construct a simple electric bell using locally available materials (iron nail, copper wire, dry cell, metal strip, wooden base). They test their bells, troubleshoot failures by applying sub-strand concepts (magnetisation, induced current, field strength), and document the design process in a project journal. Connections to the anchoring phenomenon are made explicit: the bell's electromagnet switches on and off, demonstrating controlled magnetisation and demagnetisation. Lesson 8 — Model Building, DQB Closure & Assessment: Learners produce a final annotated diagram model showing how a generator converts mechanical motion into electrical energy via electromagnetic induction, labelling all key variables that affect induced e.m.f. The DQB is reviewed: original predictions are compared with current evidence-based explanations. Learners write a structured explanation answering the driving question, complete a practical assessment of the electromagnetic induction experiment procedure, and reflect on applications of the sub-strand to Kenya's energy future.
Supporting Phenomena	– A watchman in Kisumu uses a hand-cranked torch (dynamo torch) every night. When he cranks the handle slowly the light is dim; when he cranks fast it is bright. He never replaces batteries, yet the torch always works — learners investigate why speed of rotation affects brightness. – Matatu operators in Nairobi notice that when an old speaker magnet falls onto the metal body of the vehicle, it sticks firmly, but when they heat the magnet over a jiko it loses its grip and falls off — learners investigate why heating destroys magnetism. – A maize farmer in Eldoret uses an electric fence energiser to protect her crop from wildlife. When she disconnects the coil inside the energiser, the fence stops working even though the battery is still connected — learners investigate the role of coil and magnetic core in producing current pulses. – A Standard Eight pupil in Mombasa doing a science project wraps copper wire around an iron nail and connects it to a dry cell — the nail picks up metal paper clips. When he disconnects the cell the nail drops the clips immediately. He wonders why the nail only becomes magnetic when current flows and loses magnetism the moment current stops.

LESSON 1 (80 minutes (2 × 40-minute lessons)): Anchoring Phenomenon & Methods of Magnetisation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the sub-strand by anchoring learners to the phenomenon of electromagnetic induction through a bicycle dynamo and hand-cranked torch demonstration, and to establish foundational knowledge of magnetisation methods in soft iron that will be needed to explain the phenomenon in later lessons.
Knowledge	Learners will be able to: (a) describe the methods of magnetisation of soft iron — stroking, electrical and induction methods; (b) state observations that distinguish effective from ineffective magnetisation; (c) identify soft iron as the core material in generators and dynamos.
Skills	Learners will be able to: (a) carry out the stroking, electrical and induction magnetisation experiments on soft iron nails; (b) test the strength of a produced magnet by counting the number of iron filings or small pins picked up; (c) record observations in a structured results table; (d) post 'What do we notice?' and 'What do we wonder?' questions on the Driving Question Board (DQB).
Attitudes	Learners will: (a) show curiosity and willingness to question everyday observations such as a bicycle dynamo lighting an LED; (b) appreciate that careful experimental comparison is more reliable than guessing; (c) demonstrate unity and responsibility when sharing apparatus during group investigations.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	This opening lesson plants the seed of wonder by presenting the anchoring phenomenon — a Form One student in Kericho observing a school generator and a bicycle dynamo — before learners have any formal explanation. By investigating magnetisation methods today, learners build the first piece of the puzzle: a magnet is needed before any induction can occur. Every subsequent lesson will return to the phenomenon and add another explanatory layer.
Safety Notes	1. Electrical method: use only a 1.5 V–4.5 V dry-cell battery; never connect a bare wire to the mains socket. 2. Stroking method: perform strokes away from the body to avoid scratching. 3. Handle iron nails carefully — pointed ends must face down on the bench when not in use. 4. If a hand-cranked torch is used, ensure the casing is intact with no exposed wiring. 5. Remind learners to return all magnets and nails to the tray after use to prevent misplacement or injury.

B. LESSON OVERVIEW

Learners open the sub-strand by encountering the anchoring phenomenon: a video clip and live demonstration of a bicycle dynamo lighting an LED, and a hand-cranked torch producing light without any battery. This deliberately puzzling experience is framed using the story of a Form One girl from a boarding school in Kericho who notices her school's generator and then replicates a small version with a bicycle dynamo at home. Learners write individual predictions, then share in pairs and whole class before the teacher records unresolved questions on the Driving Question Board (DQB). In the second 40-minute block, learners carry out group experiments comparing three methods of magnetising a soft iron nail — stroking, electrical (solenoid) and induction — testing the strength of each produced magnet. They record observations, discuss which method produces the strongest magnet and why, and add new "What do we wonder?" questions to the DQB. The lesson closes with learners updating a simple initial model sketch showing what they currently think is happening inside the dynamo.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners watch a 2-minute video clip (or live demonstration) showing: (1) a hand-cranked torch being cranked and lighting up, then going dark when cranking stops; (2) a bicycle dynamo wheel being spun and a small LED glowing, then going dark when the wheel stops. The teacher then reads aloud the story of the Kericho Form One student (printed on a half-page handout or projected). Learners individually write answers to three prediction prompts in their exercise books for 4 minutes: (a) 'Where do you think the electricity comes from when no battery is present?'; (b) 'Why does the LED go out the moment the wheel stops spinning?'; (c) 'What do you think is happening inside the dynamo or the generator in Kericho?'	1. Before showing the demonstration, say: 'Do NOT tell me the answer yet — I want you to watch carefully and notice everything that surprises you.' 2. Show the dynamo/torch demo twice. After the second viewing, say: 'You have 4 minutes — write your own prediction silently. No sharing yet.' Set a visible timer. 3. After 4 minutes, cold-call 4 learners (mix of high- and low-confidence learners — pre-select names): 'Amina, read me your prediction for question (b).' After each response, pause for 10 seconds WAIT TIME and ask: 'Does anyone have a different idea from Amina's?' Accept all predictions without evaluating them. 4. Say: 'We are going to keep all these ideas — right or wrong — because by Lesson 6 you will be able to explain exactly what is happening. That is our mission for this whole sub-strand.'	STRATEGY — Anchored Prediction: Requiring a written prediction before any instruction forces learners to activate prior knowledge and commit to an explanation. This creates cognitive dissonance when observations later contradict predictions, which is the engine of conceptual change. Connecting the prediction to the Kericho girl's real experience grounds the abstract physics in a familiar Kenyan boarding-school context.	Teacher circulates during the 4-minute writing period and scans 6–8 books to identify the range of ideas (e.g., 'the magnets make electricity', 'the spinning creates heat that becomes light', 'there is a hidden battery'). These misconceptions are noted mentally and will be explicitly addressed in Lessons 3–4. Observable indicator: every learner has written at least one sentence for each of the three prompts before the cold-call begins.
Phase 2 — Observe 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar	Learners work in groups of 4. Each group receives: 3 identical soft iron nails (≈ 8 cm), 1 bar magnet, 30 cm of insulated copper wire, a 1.5 V dry-cell battery with leads, a small dish of iron filings or a cluster of 10 small pins, and a results table (printed or hand-copied from the board). Groups carry out three magnetisation procedures in sequence, testing the strength of the resulting magnet after each by counting how many pins/filings the nail picks up: (A) STROKING METHOD — stroke the nail 20 times in ONE direction only using the same pole of the bar magnet; (B) ELECTRICAL METHOD —	1. Distribute apparatus and say: 'Your group has exactly 18 minutes. Assign roles now: one Materials Manager, one Recorder, one Tester, one Timer.' 2. Circulate every 3 minutes. At each group, ask one probing question — rotate questions: 'Why are you stroking in one direction only?'; 'What do you predict will happen if you stroke in both directions alternately?'; 'Where do you think the magnetism is coming from in the induction method?' Give 10 seconds WAIT TIME after each question before accepting a response. 3. If a group finishes early, prompt: 'Now stroke in BOTH directions alternately 20	STRATEGY — Comparative Experimental Investigation (NGSS Science and Engineering Practice 3: Planning and Carrying Out Investigations): By testing three methods on identical nails and measuring with the same pin-counting technique, learners generate their own comparative evidence rather than receiving a teacher explanation. The shared class results table makes patterns visible across groups and creates a data set that learners will need to explain in the next phase.	Teacher checks that each group's results table contains: (i) a pin/filing count for all three methods, (ii) a sketch of at least one method, (iii) a qualitative rating. Observable indicator: results from all groups are on the board within 20 minutes. Teacher flags any group whose induction result equals zero (nail held too far away) and asks them to repeat with the nail touching the magnet so they obtain a non-zero result.

readings	wind 15 turns of copper wire around the nail to form a solenoid, connect to the battery for 30 seconds, disconnect, then test; (C) INDUCTION METHOD — hold the nail very close to (not touching) the bar magnet for 30 seconds without touching, then test. Learners record: method used, number of pins/filings picked up, and a qualitative rating (Strong / Moderate / Weak). They also sketch the setup for each method.	times and test again. Record what happens.' 4. After 18 minutes, call time and say: 'Recorders, write your three pin-counts on the board in our shared table.' Collate results from all groups publicly.		
Phase 3 — Explain 📹 **VIDEO:** Magnetism, light and the magneto optic Kerr effect Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class discussion anchored to the board data. Learners first compare their group's results with other groups' data, noting variation and discussing reasons (e.g., number of strokes, wire turns, battery strength). The teacher then introduces the concept of magnetic domains using a simple analogy: 'Think of the iron atoms as ugali grains scattered randomly on a sufuria lid — unmagnetised. When you stroke with a magnet, it is like sweeping them all into neat rows facing the same direction.' Learners draw a before-and-after domain diagram in their books (random arrows → aligned arrows). They then connect each method to domain alignment: stroking physically aligns domains; the electrical method uses a current-produced magnetic field to align them; induction uses the external magnet's field. Learners write a 3-sentence 'Explain It' response: 'The [method] magnetises the nail because…; this connects to the dynamo/generator because…; one question I still have is…'	1. Point to the class results table and ask: 'Which method produced the most pins on average across all groups?' Cold-call 3 learners. After each answer, ask: 'What evidence from our table supports that?' WAIT 10 seconds. 2. Introduce the domain model verbally and with a quick board sketch (random arrows, then aligned arrows). Say: 'This is the same soft iron used in the core of the Kericho school generator — can anyone suggest why soft iron is chosen rather than steel?' Cold-call 2 learners. WAIT 10 seconds. Accept and validate. 3. Give 5 minutes for learners to write their 3-sentence 'Explain It' response. 4. Cold-call 3 learners to read their second sentence ('this connects to the dynamo/generator because…'). Say: 'Excellent — you have just identified the first link between magnetisation and the phenomenon we saw at the start. We will complete this connection in Lessons 2 and 3.'	STRATEGY — Analogy Bridging with Evidence Anchoring (NGSS Science and Engineering Practice 6: Constructing Explanations): The ugali-grain domain analogy gives learners a mental model grounded in a familiar Kenyan kitchen object. Requiring learners to explicitly connect the magnetisation method to the dynamo phenomenon prevents the explanation from remaining abstract and begins building the causal chain that will culminate in a full generator explanation by Lesson 6.	Teacher collects 6 'Explain It' responses (one per group) at the end of the phase and scans for: (i) correct identification of the strongest magnetisation method with a data-based reason; (ii) any attempt — however partial — to link magnetisation to the dynamo. Misconception to watch for: learners who write 'the stroking method creates electricity' — this signals a conflation of magnetisation with electromagnetic induction that must be gently corrected by redirecting: 'You have magnetised the nail — but what must happen next to produce electricity? That is what we investigate in Lesson 3.'
Phase 4 — Driving	Each learner receives two sticky notes (or small torn paper	1. Distribute notes and say: 'You have 3 minutes. Your NOTICE	STRATEGY — Driving Question Board (DQB) Launch (NGSS	Teacher photographs the DQB at the end of Lesson 1. Observable

Question Board (DQB) Creation 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	squares). On the YELLOW note they write one 'What do we NOTICE?' observation from today's lessons (phenomenon demo or experiment). On the BLUE note they write one 'What do we WONDER?' question — something today did NOT explain. Learners post their notes on the class DQB (a large sheet of paper or a section of the chalkboard divided into two columns: NOTICE | WONDER). The teacher reads 5–6 of the WONDER questions aloud and groups similar ones together in front of the class, narrating: 'These three questions are all asking about why MOVEMENT is needed — we call that cluster our movement mystery. These two are asking about what the spinning coil does — we call that our coil cluster.' The teacher writes the overarching driving question at the top of the DQB in large letters: 'How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya?'	must come from something you personally observed today — not something you already knew. Your WONDER must be a genuine question — not one you already know the answer to.' 2. As learners post notes, read a few silently. After posting, say: 'I am going to read some of your WONDER questions. Every time I read one, raise your hand if you had a similar question.' This reveals shared curiosity. 3. Cluster the questions visibly on the board. Say: 'We will track this board all the way to Lesson 6. Each lesson, we will come back and mark which questions we have now answered — in green — and add new ones.' 4. Point to the driving question and cold-call one learner: 'Wanjiku, in your own words, what is this question asking us to figure out?' WAIT 15 seconds.	Science and Engineering Practice 1: Asking Questions and Defining Problems): The DQB externalises learner curiosity, creates a public record of the class's collective knowledge gap, and gives learners agency over the inquiry. Clustering questions teaches learners that science organises problems into manageable threads. Returning to the DQB every lesson creates a visible arc of growing understanding.	indicators: (i) every learner has posted at least one NOTICE and one WONDER note; (ii) at least 60% of WONDER questions relate to the phenomenon (movement, spinning, coils, electricity without battery) rather than to magnetisation alone — this shows the phenomenon has successfully generated intellectual need.
Phase 5 — Model Building 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum]	Learners open a dedicated 'My Electromagnetic Induction Model' page in their exercise books (this page will be revised in every subsequent lesson). Today they draw an INITIAL MODEL: a simple sketch of the bicycle dynamo setup they saw in the demonstration. They must label: (a) the magnet (if they think one is present), (b) the wire or coil (if they think one is present), (c) an arrow showing the LED lighting up, and (d) a '?' symbol over any part they cannot yet explain. Below the sketch they write one sentence: 'I think electricity is produced	1. Say: 'This model is your scientific thinking on paper. It does NOT need to be correct today — it needs to be honest. The more question marks you put, the more curious you are being, and that is exactly what a scientist does.' 2. Allow 5 minutes for drawing. Circulate and place a sticky dot on 3 models that you will display — choose for variety of ideas, not correctness. 3. Facilitate the 2-minute Gallery Walk. After the walk, ask: 'What did you add a tick to that was different from your own model?' Cold-call 3 learners. WAIT 10 seconds each. 4. Close by	STRATEGY — Progressive Model Revision (NGSS Science and Engineering Practice 2: Developing and Using Models): Beginning a model on Day 1 — before instruction — establishes a baseline of learner thinking and creates a concrete artefact that learners are emotionally invested in improving. The Gallery Walk introduces collaborative model critique in a low-stakes, non-verbal format appropriate for learners who are not yet confident speaking in front of the class.	Teacher collects 4 initial model books to photograph (return next lesson). Observable indicators: (i) the sketch includes at least one labelled component (magnet OR coil OR LED); (ii) at least one '?' is present on the model, showing the learner recognises an unexplained element; (iii) the written sentence attempts a causal explanation (however naive), not just a description.

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	because…' They share their model with a partner using a 'Gallery Walk' — books open on desks, learners walk around for 2 minutes and place a tick on any model that includes an idea they had not thought of.	saying: 'In Lesson 2 we will investigate the magnetic field patterns around a magnet — that will let you upgrade the area around your magnet on this model. By Lesson 6, this model will have a complete explanation of how the Kericho school generator powers the entire dormitory.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PHENOMENON UPTAKE: Did the Kericho boarding-school story and the dynamo demonstration create genuine puzzlement? How many 'Wonder' questions on the DQB directly referenced the phenomenon (movement, spinning, coils) versus only magnetisation? If fewer than 60% referenced the phenomenon, consider re-showing the dynamo demo at the start of Lesson 2 before proceeding.

2. PREDICTION QUALITY: Did learners' written predictions reveal the key misconception that 'something inside the dynamo must be making electricity chemically'? If so, plan a brief explicit comparison of a battery (chemical → electrical) versus a dynamo (mechanical → electrical) at the start of the Explain phase in Lesson 3.

3. EXPERIMENTAL FAIRNESS: Did any groups struggle to keep the stroking method strictly unidirectional? If so, build a 30-second modelling demonstration of correct stroking technique into the start of the Observe phase in future cohorts.

4. DOMAIN MODEL ANALOGY: Did the ugali-grain analogy for magnetic domains land well across all learners? If learners in the Explain phase were writing 'the magnet puts electricity into the nail' (rather than 'aligns the domains'), the analogy needs to be revisited with a physical demonstration — scatter actual maize grains on a tray randomly, then 'sweep' them into rows with a ruler.

5. DQB MANAGEMENT: Were the question clusters you created on the DQB legible and meaningful to learners? Could learners articulate in their own words what each cluster is asking? If not, involve learners in the clustering process itself in Lesson 2 — ask them to group similar sticky notes before you name the clusters.

6. INCLUSION: Were there learners who did not post a Wonder note or whose initial model contained no labels? Follow up individually at the start of Lesson 2 with a private prompt: 'Tell me one thing from yesterday that you did not understand — let us write that as your Wonder question together.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A bicycle dynamo lights a small LED only when the wheel is spinning — the light goes out the moment the wheel stops. A hand-cranked torch behaves the same way. Inside the Kericho school generator, large copper coils rotate near magnets to power the entire dormitory — no battery or chemical reaction is involved. In our magnetisation experiments, the stroking method (20 unidirectional strokes) produced the strongest magnet on average, the electrical (solenoid) method produced a strong magnet that was demagnetised when disconnected, and the induction method produced the weakest magnet.
What did I learn?	Soft iron can be magnetised by three methods: (1) STROKING — repeated unidirectional strokes with one pole of a bar magnet physically align the magnetic domains in the iron; (2) ELECTRICAL (solenoid) — passing an electric current through a coil wound around the iron creates a magnetic field that aligns the domains (this produces an electromagnet — temporary); (3) INDUCTION — placing the iron close to a strong magnet allows the external field to align domains without contact (produces the weakest permanent magnet of the three). Magnetic domains are regions of aligned atomic magnets; an unmagnetised piece of iron has randomly oriented domains. A magnet is essential inside any generator or dynamo — without it, no electromagnetic induction can occur.

How does this explain the phenomenon?	Why is movement essential for the dynamo to produce electricity? (To be fully explained in Lessons 3–4.) Why does faster spinning make the LED brighter? (To be fully explained in Lesson 4 — factors affecting induced e.m.f.) Why does the same spinning-coil principle that lights a bicycle dynamo also power Kenya's national grid? (To be fully explained in Lesson 5–6 — applications of electromagnetic induction.) Why is soft iron used as the core material in generators rather than steel? (Partially addressed today — soft iron is easily magnetised and demagnetised; full explanation of core function in Lesson 3.)

LESSON 2 (80 minutes): Demagnetisation & Properties of Magnets: Why Stillness Kills the Current	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate three methods of demagnetisation — heating over a jiko, mechanical hammering, and electrical demagnetisation using alternating current — and use the domain theory to explain why each method destroys magnetism. They connect this understanding to the anchoring phenomenon: the Kericho student's generator coils must keep spinning because stillness — like a demagnetised bar — produces no ordered alignment and therefore no electricity.
Knowledge	Learners describe the methods of magnetisation and demagnetisation in soft iron (electrical, induction, hammering and heating) and explain observations using the concept of magnetic domains.
Skills	Learners carry out experiments on demagnetisation (electrical, hammering and heating), record and compare results, draw domain diagrams before and after each treatment, and evaluate which method is most effective and reversible.
Attitudes	Learners appreciate that understanding how magnetism is lost deepens their understanding of how it must be maintained — connecting disciplined, continuous motion in a generator to the ordered alignment of magnetic domains.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lesson 1, learners established that magnets have properties and that magnetisation creates ordered domains. In Lesson 2, learners flip the question: what destroys that order? By demagnetising samples three ways, they build the mechanistic insight that disorder in domains = no magnetism = no induced current — directly explaining why the Kericho generator coils stop producing electricity the moment they stop spinning.
Safety Notes	Jiko/charcoal heat source: use tongs at all times; keep flammable materials away; conduct heating outdoors or in a well-ventilated area. Hammering: use a rubber mallet or designated anvil block; learners wear safety goggles; no loose clothing near the work surface. Electrical demagnetisation: teacher demonstrates the AC solenoid method; learners do not handle the power supply directly. Dispose of charcoal ash safely. First-aid kit must be accessible in the laboratory.

B. LESSON OVERVIEW

Learners begin by predicting which of three demagnetisation methods will be most effective. They then carry out or observe three evidence-gathering stations — heating a magnetised nail over a jiko, hammering a magnetised iron rod on a hard surface, and watching a teacher-led AC solenoid demagnetisation — recording galvanometer/compass deflections before and after each treatment. In the Explain phase, they use domain diagrams to make sense of their results and connect domain disorder to the generator phenomenon: just as a stopped generator loses its ability to produce current, a demagnetised bar has lost its ordered domains. The DQB is updated and learners revise their cumulative models to show domains in ordered vs. disordered states.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
1 — PREDICT	Learners are shown three labelled stations: Station A (a jiko with	Display the three stations and say: 'Before we touch anything, I want	Think-Pair-Share with public prediction tally. This surfaces prior	Teacher circulates during silent writing and reads 4–5 journals.

📹 **VIDEO:** [Video: Molecule Shapes] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Earth's Magnetic Field (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	tongs and a magnetised nail), Station B (a magnetised iron rod on an anvil block with a rubber mallet), and Station C (a solenoid connected to an AC power supply — teacher-only). Before any activity begins, each learner writes a private prediction in their science journal: 'Which method do you think will most completely destroy the magnetism in the iron sample, and why?' They also predict whether the iron can be re-magnetised after each treatment. Learners share predictions with a shoulder partner (Think-Pair) and three pairs are cold-called to share with the class. Predictions are posted on a class prediction wall using sticky notes coloured by method chosen.	your brain's first answer. Write it down — there is no wrong prediction, only ones we will test.' Allow 3 minutes of silent writing. Then say: 'Turn and tell your partner which method you chose and one reason why.' Allow 90 seconds. Cold-call exactly three pairs using the class register: 'Akinyi, what did your pair predict for Station A — and why?' Apply 10 seconds of WAIT TIME before prompting. Record the three predictions verbatim on the board under the heading 'Our Predictions — We Will Return Here.' Ask the whole class: 'Raise your hand if you think heating will be most effective… hammering… electrical.' Tally results publicly. Say: 'Hold those predictions — your evidence today will either confirm or challenge them.'	conceptions about heat, mechanical shock, and electricity as disruptive forces, giving the teacher a diagnostic snapshot and giving learners a personal stake in the evidence phase.	Observable indicator: learner written predictions reference at least one causal mechanism (e.g., 'heat makes atoms vibrate and move out of line'). Prediction tally gives class-level data on misconceptions to address during Explain.
2 — OBSERVE 📹 **VIDEO:** [Video: Molecule Shapes] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Earth's Magnetic Field (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners rotate through two hands-on stations and observe one teacher demonstration, spending approximately 12 minutes at each. STATION A — Heating (jiko): Groups of 4 use a compass to confirm their nail is magnetised (deflects compass needle ≥15°). Using tongs, one learner heats the nail over the jiko charcoal for 2 minutes until glowing red. After cooling on a ceramic tile, the group re-tests with the compass and records the deflection angle. STATION B — Hammering: Groups confirm magnetisation of their iron rod with a compass. One learner (goggles on) strikes the rod firmly 20 times with a rubber mallet on the anvil block while another holds the rod pointing East–West (perpendicular to Earth's field). The group re-tests and records. STATION C —	Before rotation begins, say: 'At every station your job is to be a scientist in Kericho — you are gathering evidence, not guessing. Record numbers, not just words.' Model how to read the compass deflection angle. At Station A, circulate and ask: 'What do you notice happening to the compass needle as the nail cools? Do not explain yet — just describe what you see.' Apply 12 seconds of WAIT TIME. At Station B, ask: 'What does each hammer strike feel like — is energy going in or out of the rod?' At Station C (teacher demonstration), narrate aloud: 'Watch the galvanometer as I slowly withdraw the rod from the solenoid — tell me the moment you see any change.' Cold-call two learners to read the galvanometer values. After all rotations, say: 'Look at your results table. Without	Structured multi-station evidence gathering with a shared results table. Requiring numerical deflection readings (not just 'magnetism went away') forces learners to compare effect sizes across methods and prepares them to use evidence quantitatively in the Explain phase.	Teacher checks results tables during rotation. Observable indicator: all four cells of the results table are completed with numeric compass readings for each station. Groups that record only qualitative terms ('it stopped working') are prompted: 'Give me a number — what angle is the needle at now compared to before?'

	Electrical (teacher demonstration): Teacher passes the magnetised rod slowly out through an AC solenoid (connected to 6V AC supply) three times while learners observe the galvanometer reading drop to zero. Learners sketch the solenoid setup and record the before/after compass deflection read aloud by teacher. All groups complete a results table: Method | Compass deflection before | Compass deflection after | Observation.	explaining yet, which station produced the biggest change in compass deflection? Discuss in your group for 60 seconds.' Cold-call three groups using the register.		
3 — EXPLAIN 📹 **VIDEO:** Video: Molecule Shapes Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Earth's Magnetic Field (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher reveals a large domain diagram on the board showing two states: (i) ordered domains (magnetised bar — all arrows pointing the same direction) and (ii) disordered domains (demagnetised bar — arrows pointing randomly). Learners are given a blank domain diagram worksheet with three bars labelled A, B, and C. Working in pairs, they draw the domain state BEFORE and AFTER each demagnetisation method, labelling what happened to the domains in each case. Pairs write one sentence per method explaining the mechanism (e.g., 'Heating gives domain walls enough energy to break from their aligned positions and point randomly'). Three pairs share their diagrams under the document camera. The class then connects this to the phenomenon: learners answer in writing — 'The Kericho student holds her bicycle dynamo wheel perfectly still. Using domain language, explain what is happening inside the generator and why no current flows.' Learners also compare: which method is reversible (the sample	Reveal the domain diagrams and say: 'This is the language scientists use to explain what you just observed. Your job now is to use it.' Distribute worksheets. After 8 minutes of pair work, say: 'I am going to pick three pairs to share their Station A, B, and C diagrams — be ready to defend your arrows.' Cold-call by register. After each sharing, ask the class: 'Do you agree with the direction of the arrows after demagnetisation? Thumbs up, thumbs sideways, thumbs down.' Apply 10 seconds of WAIT TIME before taking hands. For the phenomenon connection question, give 4 minutes of silent individual writing. Then cold-call two learners: 'Wanjiku, read your sentence to us.' Follow up: 'Can anyone add to what Wanjiku said using the word domain?' For the reversibility discussion, pose: 'If you were the engineer maintaining the Kericho generator, which demagnetisation risk would worry you most — accidental heat, vibration, or electrical interference? Why?' Allow 90 seconds of pair talk then cold-call one pair.	Domain diagram annotation (Representational Mapping strategy). By physically drawing arrows for each method, learners translate their observational evidence into a mechanistic model — the core sensemaking move that connects macro observations (compass deflection drops) to micro cause (domain disorder). The phenomenon connection question then transfers this mechanism to the generator context.	Collect domain diagram worksheets at the end of the phase. Observable indicators: (1) Before-state diagrams show aligned arrows for all three bars. (2) After-state diagrams show randomised arrows. (3) Written mechanism sentences include at least one causal term (energy, vibration, alternating field, random). (4) Phenomenon connection response uses the word 'domain' and references movement/stillness.

	can be re-magnetised) and which is most destructive to the material itself.			
4 — DRIVING QUESTION BOARD (DQB) UPDATE 📹 **VIDEO:** [Video: Molecule Shapes](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Earth's Magnetic Field (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board from Lesson 1. Individually, each learner writes on a green sticky note one thing they can NOW answer or partially answer about the driving question ('How does moving a wire through a magnetic field produce electricity…?') as a result of today's lesson. They write on a yellow sticky note one NEW question today's lesson has raised for them. Learners place their notes on the DQB and do a 3-minute gallery walk to read three peers' new questions. The class votes (show of hands) on the top two new questions to prioritise in upcoming lessons. The teacher records these on the DQB anchor chart.	Say: 'Our driving question is still open — but today you have new evidence. Green note: what can you now answer that you could not answer at the start of Lesson 1? Yellow note: what new question has today's evidence raised for you?' Allow 3 minutes of silent writing. During gallery walk, say: 'Read three notes that are not yours. Place a small dot sticker next to any question you also want answered.' Tally dot stickers and read the top two questions aloud: 'These are YOUR questions — we will pursue them in Lessons 3 and 4.' Point to the phenomenon description posted on the wall: 'Notice that the Kericho student asked: why does faster spinning make the light brighter? Today's lesson gives us part of that answer. What part?' Cold-call one learner. Add their response as a class annotation on the phenomenon card.	DQB iterative update (Claim-Evidence-Question cycle). Green notes require learners to articulate a knowledge gain explicitly linked to evidence; yellow notes sustain productive intellectual discomfort and drive the inquiry sequence forward. Public dot-voting builds shared ownership of the investigation.	Teacher reads all green notes before the next lesson. Observable indicator: green notes reference domain order/disorder, demagnetisation, or the generator phenomenon — not generic statements like 'I learned about magnets.' Yellow notes that reveal persistent misconceptions (e.g., 'Does the magnet get used up?') are flagged for targeted address in Lesson 3.
5 — MODEL BUILDING 📹 **VIDEO:** [Video: Molecule Shapes](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Earth's Magnetic Field (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative model from Lesson 1 (a sketch of the Kericho generator with rotating coils near a magnet). They now add a second panel labelled 'Domain View — Magnetised vs. Demagnetised.' In the new panel they draw: (a) the magnet in the generator with aligned domain arrows, and (b) a hypothetical 'stopped and heated magnet' with disordered domain arrows and a label showing zero current output. Using a coloured pen, they annotate the original generator sketch with the phrase 'Domains MUST stay ordered for current to	Distribute learners' Lesson 1 models and say: 'Today you are adding a new layer — the domain view. Your model must now show what is happening inside the magnet, not just around it.' Circulate and prompt: 'Your domain arrows in the magnetised bar — are they all pointing the same direction? Good. Now in the demagnetised bar — are they random? That is the key difference.' After 10 minutes, ask three volunteers to hold up their models under the document camera and explain one addition they made. Ask the class: 'Which	Cumulative model revision (Progressive Modelling strategy). Adding a domain-level panel forces learners to connect macroscopic observations (compass deflection, light brightness) to a sub-microscopic explanatory mechanism. The Demagnetisation Risk Register bridges school science to authentic engineering practice in the Kenyan energy sector.	Teacher reviews 6–8 models before Lesson 3 using a 3-point check: (1) Domain arrows correctly ordered in magnetised state ✓, (2) Domain arrows correctly disordered in demagnetised state ✓, (3) At least one annotation explicitly linking domain order to the generator/current production ✓. Models missing all three checks receive a written prompt: 'What do the arrows inside the magnet look like when the generator is running? Add this to your model.'

	flow — motion maintains the condition for induction.' Learners also add a 'Demagnetisation Risk Register' box listing the three risks (heat, mechanical shock, AC interference) and one real-world engineering precaution for each (e.g., 'Generator housings are cooled to prevent heat demagnetisation'). Models are signed, dated, and stored in individual science portfolios for revision in subsequent lessons.	model shows the domain concept most clearly — and why?' Apply 12 seconds of WAIT TIME. Close by saying: 'Every time you add to this model, you are building the same understanding that Kenyan power engineers at Kenya Power use when they maintain the generators supplying the national grid. Your model is real engineering thinking.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) Were learners able to transfer the domain model from the demagnetisation experiments to the generator phenomenon, or did they treat them as two separate topics? What bridging question could you use next time? (2) Which of the three demagnetisation stations generated the richest discussion — and why? Was it the one with the most dramatic result (likely heating) or the one with the most surprising result? (3) Review the green DQB notes: did learners' stated knowledge gains reflect mechanistic understanding (domain disorder) or surface description (magnet got weaker)? Plan one targeted question for the start of Lesson 3 to probe the depth of mechanistic understanding. (4) Consider equity: did all groups — including quieter learners and girls — have equal access to the hands-on roles at each station? If not, how will you rotate roles more deliberately in Lesson 3? (5) The phenomenon connection to Kenya Power and the national grid is a recurring thread — did you make this connection explicit enough, or did it feel forced? Adjust the phrasing for Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that a magnetised nail lost its ability to deflect a compass needle after being heated to red-hot over a jiko; a magnetised iron rod lost its deflection after 20 hammer strikes on an anvil block; and a magnetised rod showed zero galvanometer response after being slowly withdrawn through an AC solenoid — demonstrating three distinct pathways to demagnetisation.
What did I learn?	Learners learned that magnetism depends on the ordered alignment of magnetic domains; any process that supplies enough energy (heat, mechanical shock) or repeatedly reverses the field direction (alternating current) randomises the domains and destroys magnetism. This domain disorder is analogous to why a stationary generator produces no current — without continuous rotation, the conditions for ordered induction are not maintained.
How does this explain the phenomenon?	Learners explained that the Kericho student's bicycle dynamo light goes out when the wheel stops because stopping the wheel is functionally equivalent to demagnetisation — the ordered relationship between the magnetic field and the moving conductor collapses, just as ordered domains collapse when a magnet is heated or hammered. Engineers at Kenya Power protect generator magnets from heat and vibration for exactly this reason: domain disorder means power failure.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Magnetic Field Patterns: Mapping the Invisible Force	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to map, draw, and interpret magnetic field patterns around bar magnets and U-shaped magnets, and to connect field concentration at poles to the operating principle of the generator observed in the Kericho boarding school phenomenon.
Knowledge	Learners will be able to: (a) describe the magnetic field patterns around a bar magnet and a U-shaped magnet; (b) explain field direction conventions (North pole to South pole outside the magnet); (c) describe the significance of a uniform magnetic field between the poles of a U-shaped magnet; (d) relate field strength (line density) to pole regions and explain why this matters for electromagnetic induction.
Skills	Learners will be able to: (i) use iron filings and plotting compasses to experimentally map magnetic field lines around bar and U-shaped magnets; (ii) draw accurate, annotated field diagrams with correct direction arrows; (iii) analyse field diagrams to identify regions of strong and weak field; (iv) record and communicate observations using labelled scientific diagrams.
Attitudes	Learners will: appreciate that invisible physical phenomena (magnetic fields) can be made visible through evidence-gathering; value careful, systematic observation as a foundation for scientific explanation; show curiosity about how the structure of a U-shaped magnet's field directly enables electricity generation in Kenya's power stations.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lesson 1 learners established that movement is essential to the dynamo and generator. In Lesson 2 they investigated magnetisation and demagnetisation to understand how a magnet is created and maintained. Now in Lesson 3 learners make the invisible magnetic field visible — they map field patterns, discover that field lines crowd together at the poles, and recognise that the dense, uniform field between the poles of a U-shaped magnet is precisely the region through which a wire must move to generate the largest possible induced e.m.f. This sets up the critical question for Lessons 4–5: what happens when a conductor enters that concentrated field region?
Safety Notes	Iron filings must be kept away from eyes and clothing — learners should handle the trays carefully and avoid blowing on the filings. Magnets must be stored in keeper pairs after use to prevent demagnetisation. Learners with pacemakers or metallic implants should inform the teacher before handling strong magnets. Sweep up spilled iron filings immediately to avoid ingestion or eye contact. Plotting compasses should not be placed on top of each other or near electronic devices.

B. LESSON OVERVIEW

Learners begin by predicting what the magnetic field around a bar magnet and a U-shaped magnet looks like, sketching their ideas before any equipment is introduced. They then conduct a structured practical investigation — first sprinkling iron filings onto paper placed over each magnet type, then using plotting compasses to trace field direction step-by-step — producing two annotated field diagrams. During the Explain phase the teacher facilitates sensemaking around three key ideas: (1) field lines run from North to South outside the magnet, (2) line density indicates field strength (poles are strongest), and (3) the field between the poles of a U-shaped magnet is uniform and intense — exactly the condition needed for maximum induction in a generator. Learners update the Driving Question Board with new evidence-linked questions and revise their cumulative generator model to show where the magnetic field exists and why its shape matters. Throughout, the lesson is anchored to the Kericho school generator and the bicycle dynamo: the copper coils of the school generator rotate inside a housing that functions exactly like a large U-shaped magnet, and the dense uniform field between those poles is what makes the dormitory lights shine.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict 📹 **VIDEO:** Representing quantities with vectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Earth's Magnetic Field (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a blank A4 sheet showing an outline of a bar magnet (labelled N and S) and a separate outline of a U-shaped magnet (poles facing each other). Without any equipment, learners individually sketch what they think the magnetic field looks like around and between each magnet — drawing lines, arrows, and annotating where they think the field is strongest. They write a one-sentence justification below each sketch: 'I think the field is strongest at _______ because _______.' After 4 minutes of individual thinking, learners share predictions in pairs (Think-Pair-Share), then three pairs are cold-called to present contrasting predictions to the class. Predictions are pinned to a designated 'Before Evidence' section of the classroom wall for comparison after the investigation.	Distribute prediction sheets face-down. Say: 'Before we touch a single magnet today, I want to know what you already think the invisible field around a magnet looks like. Flip your sheet over — you have 4 minutes, working completely alone. Draw what you imagine, and tell me in one sentence where you think the field is strongest and why.' [Set a visible timer for 4 minutes. Circulate silently — do NOT correct or affirm.] After 4 minutes: 'Now turn to your elbow partner. You have 90 seconds — compare your sketches. Do you agree on where the field is strongest?' [Allow 90 seconds.] Cold-call exactly 3 pairs using name sticks — one pair who drew lines curving around the bar magnet, one who drew straight lines, one who left the U-shaped magnet region blank. Ask each: 'Walk us through your thinking — why did you draw it that way?' [Allow 10-second WAIT TIME after each response before probing.] Pin all prediction sheets to the wall. Say: 'Hold onto these ideas — your job this lesson is to find evidence that confirms, refines, or challenges what you have predicted.'	Think-Pair-Share with prediction pinning. This strategy activates prior knowledge from Lesson 2 (magnetisation and poles) and creates cognitive commitment to a specific prediction — learners are then motivated to test their own ideas during the investigation. Pinning predictions to the wall makes thinking visible and creates a reference point for later revision, reinforcing the science practice of engaging in argument from evidence.	Observable indicator: Do learners' sketches show any understanding of field directionality (N to S) or field concentration at poles from prior learning? Teacher looks for: (a) whether learners draw closed or open field lines, (b) whether arrows are present and, if so, whether they point toward or away from the North pole, (c) whether any learner correctly predicts the uniform parallel-line pattern between U-shaped magnet poles. These baseline data inform how much scaffolding to provide in Phase 2.
Phase 2 — Observe 📹 **VIDEO:** Representing quantities with vectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners work in pre-assigned groups of four. Each group receives: one bar magnet, one U-shaped magnet, iron filings in a small dispensing shaker, two sheets of white A4 card, a plotting compass, a ruler, and pencils. Activity A (10 min): Place the A4 card over the bar magnet. Gently	Before distributing equipment: 'There are iron filings in those shakers. They are fine metal particles — do NOT blow on them, do NOT tip the shaker upside down, and do NOT touch your eyes until you have washed your hands. If filings spill on the desk, use the small brush in your tray to	Structured Inquiry with Annotated Diagram Production (NGSS Science and Engineering Practice 3: Planning and Carrying Out Investigations; Practice 4: Analysing and Interpreting Data). Learners do not simply observe — they produce a scientific artefact (the annotated field diagram) that	Observable indicators: (a) Field line arrows point from North to South pole outside the magnet on both diagrams — teacher checks at least 2 groups' diagrams during circulation. (b) Learners annotate the region between U-shaped magnet poles as 'uniform' or equivalent — if fewer than half the

for similar videos 📖 **HTML:** [Earth's Magnetic Field (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	tap and shake iron filings from the shaker onto the card. Observe the pattern that forms, then carefully trace the field lines onto the card with a pencil while the filings are still in place. Add arrows using the plotting compass to determine direction (compass N points toward which pole outside the magnet?). Label the diagram: poles, field lines, arrows, and two annotations — 'strongest field' and 'weakest field'. Activity B (10 min): Repeat with the U-shaped magnet. Focus especially on the region between the two poles. Learners annotate: 'uniform field region' and 'crowded field lines at poles'. Activity C (5 min): Each group completes a structured observation table with three columns — 'What we saw (bar magnet)', 'What we saw (U-shaped magnet)', 'One difference between the two patterns'. Groups share one key difference in a whole-class round-robin (30 seconds per group, no repeats allowed).	sweep them back — do not wipe with your hand.' [Pause 5 seconds, make eye contact with every group.] As groups work, circulate and use targeted prompts: To groups who see the filings pattern but have not added arrows — 'Your plotting compass needle has a red end. What does that red end do when you put it near the North pole of the magnet? Use that to decide which way your arrow should point.' [10-second WAIT TIME.] To groups working on the U-shaped magnet — 'Look at the lines between the poles. What word would you use to describe them compared to the lines at the sides of the bar magnet? Write that word on your diagram.' To fast-finishing groups — 'Now hold your two completed diagrams side by side. If you were a tiny charged particle moving through each field, where would you experience the strongest push? Mark that on both diagrams with a star.' For the round-robin: 'Each group has 30 seconds — give me ONE difference between the bar magnet pattern and the U-shaped magnet pattern that has not already been said. I am timing you.' [Use a stopwatch visibly.]	encodes their observation. The structured observation table forces comparison between the two magnet types, surfacing the critical insight that the U-shaped magnet creates a confined, uniform, intense field — the property that makes it useful in a generator. The round-robin share enforces active listening and prevents free-riding.	groups do this unprompted, teacher pauses the class and asks: 'Everyone freeze — look at the lines between the poles of your U-shaped magnet. Are they curved or straight? Parallel or spreading out? Add one word to that region now.' (c) Observation tables contain at least one scientifically accurate difference between the two patterns.
Phase 3 — Explain 📹 **VIDEO:** [Representing quantities with vectors] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Earth's Magnetic	Learners remain in groups but now work through a structured explanation sequence. Step 1 — Gallery Walk (5 min): Groups post their two annotated field diagrams on the wall. Learners do a silent gallery walk with a sticky note, placing a green dot on one diagram from another group that shows something they agree with, and writing one question on a yellow sticky note for any diagram that confuses them. Step 2 —	Launch gallery walk: 'You have 5 minutes — silent gallery walk. Green dot on one diagram you think is scientifically correct. Yellow sticky question on any diagram you find confusing. Go.' [Timer visible. Circulate and read sticky notes to identify misconceptions for the whole-class discussion.] For whole-class sensemaking, draw a large bar magnet field diagram on the board and ask: 'Somebody described the lines	Gallery Walk with Sticky-Note Critique followed by Phenomenon-Linked Writing (NGSS Practice 6: Constructing Explanations; Practice 8: Obtaining, Evaluating, and Communicating Information). The gallery walk turns peer work into evidence to be evaluated, not just admired. The three-sentence Phenomenon Link forces learners to transfer the abstract concept (uniform field, field density) into the specific context of the Kericho	Observable indicators: (a) In gallery walk sticky notes — are yellow questions about field direction or field density (productive confusion) or about completely unrelated things (lost learners requiring re-teaching)? (b) In whole-class exchange — can cold-called learners articulate the link between line density and field strength without being told? If not, teacher uses analogy: 'Imagine traffic on Nairobi's Mombasa Road

Field (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class sensemaking (10 min): Teacher displays a large reference field diagram on the board (bar magnet and U-shaped magnet side by side) and facilitates explanation of three key concepts through structured questioning: (i) Field direction convention — lines go from N to S outside the magnet; (ii) Field line density = field strength — crowded lines at poles mean strong field; (iii) The uniform field between U-shaped magnet poles — parallel, equally spaced lines showing a constant field direction and magnitude. Step 3 — Connection to phenomenon (5 min): Learners individually write a 3-sentence 'Phenomenon Link' in their science notebooks: 'The Kericho generator works like a large U-shaped magnet because ______. The copper coils rotate in the region between the poles because ______. If the poles were farther apart or the field was weaker, the generator would ______.' Groups compare sentences and refine.	between the U-shaped magnet poles as parallel and equally spaced. What does equally spaced lines tell us about the strength of the field across that whole region?' [10-second WAIT TIME. Cold-call a learner who has not spoken yet.] After response: 'Exactly — or let's refine that together. If the lines are equally spaced, the field is the same strength everywhere in that region. We call that a uniform field. Write those two words on your diagram now: UNIFORM FIELD.' Then hold up a picture of the Kericho generator coils (or sketch one on the board): 'Look at this. This is the inside of the generator at the school in Kericho. The copper coils rotate — where do they rotate?' [Point to the gap between the poles.] 'They rotate exactly in the region we just described as uniform and strong. Why would an engineer design it that way?' [10-second WAIT TIME. Cold-call 2 learners.] For Phenomenon Link sentences: 'I am going to give you 3 minutes — alone, in your notebook — to write three sentences connecting today's field patterns to the Kericho generator. Start with the sentence starters on the board.' [Display starters. Timer 3 minutes. Then: 'Share with your group — does every sentence contain a scientific term we used today?']	generator, making the explanation functional and not merely definitional. This directly addresses the KICD key inquiry question: how is the study of electromagnetic induction important in daily life?	at rush hour — cars packed tightly together versus widely spaced on a rural road in Turkana. Which road has more vehicles per metre? That density is like field line density.' (c) Phenomenon Link sentences — teacher reads 4–5 notebooks during the group-share phase; sentences must include at least one of: 'uniform field', 'strong field at poles', 'concentrated field lines', 'between the poles'.
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Representing quantities with vectors	Each learner receives two sticky notes (one blue = 'I now understand', one orange = 'New question I have'). On the blue note they write one thing from today's lesson that helps answer the main driving question: 'How does moving a wire through a magnetic field produce electricity?' On the	Distribute sticky notes. Say: 'Blue note — one thing you now understand that helps explain why the Kericho generator works or why the bicycle dynamo lights up. Orange note — one new question today's evidence has raised for you. You have 2 minutes. Go.' [Timer. Circulate — prompt stuck	Driving Question Board Curation (NGSS Practice 1: Asking Questions; Practice 7: Engaging in Argument from Evidence). The DQB is not a passive display — it is actively managed. The two-colour sticky note system separates consolidation from curiosity, and the public	Observable indicators: (a) Blue notes contain specific scientific language from today (field lines, poles, uniform field, field direction, U-shaped magnet) — not vague statements like 'magnets are strong'. (b) Orange questions are evidence-based and forward-looking — they reference

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Earth's Magnetic Field (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	orange note they write a new question that today's evidence has raised — specifically prompted by the lesson's sub-question: 'Where is the field strongest — and why does that matter for a generator?' Learners post notes on the DQB under the columns: 'What We Now Know' (blue) and 'New Questions' (orange). The class does a 3-minute standing review of the board — teacher reads 3–4 orange questions aloud, categorises them on the board under 'We will answer this in Lesson 4', 'We will answer this later in the unit', or 'This connects to something outside this unit'. Learners update the running 'Evidence We Have Gathered' column of the DQB with today's key findings: (1) Magnetic fields have direction (N→S outside magnet). (2) Field is strongest where lines are densest — at the poles. (3) U-shaped magnets create a uniform, intense field between their poles. (4) The Kericho generator coils rotate in this uniform field region.	learners: 'Think about the uniform field we described. What do you still not know about what happens when the wire actually moves through that field?'] After posting: Stand at the DQB and read 3 orange notes aloud — choose ones that preview Lesson 4 content (e.g., 'What happens to the galvanometer when we move the wire through the uniform field?' or 'Does the direction of movement matter?'). Say: 'This orange note asks — [read it]. Class, is that something we can answer today, or do we need more evidence?' [Cold-call 2 learners for responses. Accept all reasonable answers.] 'We will return to this exact question in our next lesson. Keep it visible on the board.' Update the 'Evidence' column together by asking: 'What are the three most important things the iron filings and plotting compass showed us today? Somebody give me number one.' [Cold-call. Write response in column. Repeat for 2 and 3.]	categorisation of new questions by the teacher models how scientists identify what is known versus what requires further investigation. This directly scaffolds the evidence-gathering sequence across all 8 lessons.	movement, conductors, or electricity generation, showing that learners are mentally connecting today's field mapping to the upcoming induction experiments. (c) If more than a third of orange notes are still asking questions that were answered in today's lesson (e.g., 'Why do field lines curve?'), teacher recognises incomplete understanding and flags those concepts for a brief revisit at the start of Lesson 4.
Phase 5 — Model Building 📹 **VIDEO:** Representing quantities with vectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Earth's Magnetic Field (Flexbook) Source: CK-12 🔍 Search ARES	Learners retrieve their cumulative generator model diagram begun in Lesson 1 and revised in Lesson 2. Today they add a new layer to the model: the magnetic field. Using the annotated field diagrams produced in Phase 2 as a reference, each learner: (1) Draws the field lines and direction arrows around the magnet component of their model — showing the field in the region where the coil will rotate. (2) Highlights or shades the 'uniform field zone' between the poles of the U-shaped magnet component. (3) Adds a written annotation: 'The coil must rotate in	Say: 'Take out the generator model you have been building since Lesson 1. Today you are adding the invisible — you are adding the magnetic field to your model. Use your field diagrams from the investigation as your reference. You have 6 minutes to add field lines, label the uniform field zone, and write why the coil must rotate in that zone.' [Timer. Circulate — prompt learners who only draw field lines without connecting to the coil: 'Where exactly does the coil of the Kericho generator sit relative to these field lines? Draw an arrow showing the	Cumulative Model Revision with Annotated Limitation Notes (NGSS Practice 2: Developing and Using Models). The model is not redrawn from scratch — it is iteratively revised, which mirrors authentic scientific modelling practice. The 'Model Limitation' note is a metacognitive tool: it forces learners to identify the boundary of their current understanding and transforms that boundary into a productive question. This directly links to the storyline: the model will only be complete after Lessons 4 and 5 (electromagnetic induction experiments), so each limitation	Observable indicators: (a) Field lines in models show correct direction (N to S outside magnet) and crowding at poles — teacher verifies at least 6 individual models during the 6-minute drawing phase. (b) The 'uniform field zone' is drawn between the poles of the U-shaped magnet component, not at the sides. (c) The annotation 'The coil must rotate in this zone because ______' contains a causal claim involving field strength or field density — not just 'because that's where it fits'. (d) Model Limitation notes are specific and forward-looking (e.g., 'My

for similar readings	this zone because ______.' (4) Writes a 'Model Limitation' note: 'My model does not yet show ______.' Groups of four compare models for 3 minutes — identifying one thing that is the same across all four models (class consensus) and one thing that differs (productive disagreement to be resolved with future evidence). Groups nominate a 'model reporter' who shares the disagreement with the class in 30 seconds.	coil's position.'] After 6 minutes: 'Now compare with your group — 3 minutes. Find one thing all four of your models agree on and one thing where your models look different. You are not trying to decide who is right — you are identifying what evidence you would need to settle the disagreement.' For model reporters: 'One disagreement per group — 30 seconds, go.' [After each report, ask the class: 'Does anyone have evidence from today that settles this disagreement?' Cold-call 1 learner per report.] Close the phase: 'Write your Model Limitation note now — what is the one thing your model cannot yet explain? This is your personal research question going into Lesson 4.'	note is a prediction about what future evidence will reveal.	model does not yet show what happens to current when the wire moves', 'My model does not show why faster spinning gives more electricity') rather than trivially incomplete (e.g., 'My model does not show colours').

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PREDICTION QUALITY: When you pinned learners' prediction sheets to the wall, how many correctly predicted the shape of the uniform field between the U-shaped magnet poles before the investigation? What does this tell you about the prior knowledge learners bring from Junior School Integrated Science — and how will you use this baseline in Lesson 4?

2. IRON FILINGS MANAGEMENT: Did any group struggle with the iron filings technique — getting a clear pattern, or avoiding filings moving before they could trace them? If so, note which groups and plan a brief re-demonstration at the start of Lesson 4 using a visualiser or document camera so the technique is visible to all.

3. PHENOMENON CONNECTION: During the Explain phase, when you asked 'Why would an engineer design the Kericho generator with coils rotating in the uniform field region?' — how many learners were able to answer with field-strength reasoning without being told? If fewer than half could, consider adding a bridging question at the start of Lesson 4: 'If I want to push a wire as hard as possible with a magnetic force, do I want to put it where the field is weak or where it is strong? How does that connect to the generator?'

4. DQB ORANGE NOTES: Review the orange question sticky notes before Lesson 4. Categorise learner questions into: (a) questions about conductor movement in a field (these are the target for Lesson 4), (b) questions about current direction and factors affecting e.m.f. (target for Lessons 5–6), (c) questions about generator applications in Kenya (target for Lesson 7–8). Use the Lesson 4 category questions as your explicit launch hook for that lesson.

5. MODEL LIMITATION NOTES: Collect or photograph learners' Model Limitation notes. These are your best diagnostic of where individual learners are in the storyline. Any learner whose limitation note does not reference movement, current, or induction may need targeted support before Lesson 4's induction experiment.

6. EQUITY CHECK: Did female learners have equitable access to the plotting compass and hands-on iron filings work, or did male learners dominate the physical manipulation? If needed, assign explicit roles in Lesson 4: the learner who held the compass today must be the one who moves the conductor through the field in the next experiment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners sprinkled iron filings over bar magnets and U-shaped magnets and used plotting compasses to trace the direction of magnetic field lines, producing two annotated field diagrams showing field line patterns, direction arrows (North to South outside the magnet), crowding of lines at poles, and the uniform parallel-line pattern between the poles of the U-shaped magnet.
What did I learn?	Magnetic fields have direction (field lines run from the North pole to the South pole outside a magnet); field line density indicates field strength — lines are most crowded at the poles where the field is strongest; the U-shaped magnet produces a uniform, intense magnetic field in the region between its poles; this uniform field region is exactly where the copper coils of the Kericho school generator (and all generators) rotate, because the strong, constant field in that zone maximises the force on the moving conductor.
How does this explain the phenomenon?	The Kericho boarding school generator works because the large copper coils rotate inside a housing that functions like a giant U-shaped magnet — the coils spin through the intense, uniform field between the poles. The bicycle dynamo that the student removed from the old bike uses the same principle on a smaller scale. Today's lesson makes the invisible field visible: where field lines are densest, the magnetic force is greatest, and it is in that dense-field zone that movement of a conductor will produce the largest induced e.m.f. — which is why faster spinning of the dynamo wheel produces a brighter LED light.

LESSON 4 (80 minutes): Introducing Induced E.M.F (Qualitative): What Makes the Galvanometer Deflect?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out the KICD-prescribed electromagnetic induction experiment using a U-shaped magnet, a sensitive galvanometer and a straight copper conductor to observe how conductor motion relative to a magnetic field induces an electromotive force (e.m.f).
Knowledge	Learners describe induced electromotive force (e.m.f) in electromagnetism and identify the conditions under which it is produced, including the role of relative motion between a conductor and a magnetic field.
Skills	Learners perform an experiment on electromagnetic induction, recording galvanometer deflection direction and magnitude for four conductor positions: (a) stationary between poles, (b) moved parallel to the field, (c) moved vertically upward, (d) moved at an angle to the field.
Attitudes	Learners appreciate the significance of induced e.m.f as the foundational principle linking the Kericho school generator and the bicycle dynamo phenomenon — recognising that movement, not chemical reaction, is the source of electrical energy.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lessons 1–3 learners established that magnets have fields, that conductors can be magnetised, and that field patterns exert force on charges. In Lesson 4, learners cross the critical threshold: they discover that MOVING a conductor through a field forces current to flow — directly explaining why the Kericho student's dynamo wheel must keep spinning for the LED to stay lit, and why the generator coils must rotate continuously to supply the dormitory with electricity.
Safety Notes	Ensure galvanometer terminals are not short-circuited. Handle the U-shaped magnet with care — do not drop it or place it near electronic devices, other magnets, or iron-containing objects on the desk. Connecting wires must have intact insulation. Learners must not touch bare wire ends during the experiment. Wash hands after handling laboratory apparatus.

B. LESSON OVERVIEW

Learners open with a prediction task — sketching what they expect the galvanometer needle to do in each of four conductor positions. They then carry out the KICD-prescribed electromagnetic induction experiment in groups of four, recording deflection direction and magnitude in a structured results table. A whole-class data collation on the board surfaces the pattern: deflection occurs only when there is relative motion between conductor and field, and its magnitude depends on the angle and speed of movement. Learners use a Claim–Evidence–Reasoning (CER) frame to construct a first written explanation of induced e.m.f, explicitly linking their findings to the Kericho bicycle dynamo and the school generator. The lesson closes with DQB revision and a cumulative model update, advancing the class storyline toward Lesson 5 (factors affecting induced e.m.f — quantitative).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict	Each learner receives a half-page PREDICT SHEET showing a diagram of the U-shaped magnet	Distribute PREDICT SHEETS face-down. Say: 'Before we touch any equipment today, I want you to	Think-Pair-Share Prediction Elicitation — activates prior knowledge from Lessons 1–3 (field	Teacher scans predict sheets during the 4-minute window and notes how many learners predict

📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	with a straight conductor between its poles and four labelled scenarios: (a) conductor stationary, (b) conductor moved parallel to field lines (horizontal), (c) conductor moved vertically upward between poles, (d) conductor moved at 45° to field. Learners individually sketch a galvanometer needle position for each scenario — deflected left, deflected right, or centre — and write a one-sentence reason. Learners then share predictions with a shoulder partner (Think-Pair) before the teacher cold-calls three pairs to share contrasting predictions with the class.	be scientists making a prediction. Turn your sheet over — you have 4 minutes to sketch the galvanometer needle for each scenario and write ONE reason. Work alone.' Start a visible 4-minute timer. After 4 minutes, say: 'Now share your prediction for scenario (c) — the conductor moving upward — with your shoulder partner. You have 90 seconds.' After sharing, cold-call 3 pairs: 'Amina and Korir — what did you predict and why? ... Otieno and Wanjiru — did you agree or disagree? ... Halima and Muthoni — what was the key disagreement?' Accept all predictions without correcting. Write contrasting predictions on the board in two columns labelled 'Deflection' and 'No Deflection'. Say: 'We will test every one of these predictions in the next 25 minutes. Keep your predict sheet — you will need it at the end of the lesson.'	patterns, force on conductors) and creates cognitive dissonance between learners who expect deflection when stationary and those who do not, motivating close observation during the experiment.	deflection for scenario (a) — stationary conductor. Learners who predict deflection in scenario (a) hold an incomplete model of induction and will need targeted questioning during the observe phase. Observable indicator: learner writes a reason that references either 'field', 'motion', 'wire', or 'current' — any of these shows engagement with prior learning.
Phase 2 — Observe 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four collect apparatus: 1× U-shaped (horseshoe) magnet, 1× sensitive centre-zero galvanometer, 1× 30 cm straight copper conductor with alligator-clip leads, connecting wires. Each group follows a printed PROCEDURE CARD (aligned to the KICD assessment rubric) with four steps: (a) Lay the conductor stationary between the poles — hold it still for 10 seconds — read galvanometer. (b) Move conductor slowly parallel to field lines (left-right, horizontal) — read galvanometer. (c) Move conductor vertically upward between poles at moderate speed — read	Before groups start, demonstrate apparatus assembly at the front bench: 'Connect the two ends of the copper conductor to the galvanometer terminals using the clip leads — like this. The galvanometer should read zero when the wire is still and away from the magnet. Check this now before you begin.' Circulate during the experiment. At each group, use these specific prompts: — 'What is the needle doing RIGHT NOW? Is it moving or still?' — 'What are you doing differently between step (a) and step (c)?' — 'Try moving the conductor faster in step (c). What changes?' If a group gets no deflection in	Structured Data Collation (Class Data Wall) — pooling results from all groups makes the pattern statistically clear even if individual group measurements contain small errors. Publicly displaying group data creates accountability and allows anomalies to be identified and discussed rather than ignored.	Teacher checks results tables during circulation — the critical observable indicator is whether learners record CENTRE (zero) for scenario (a) — stationary conductor. Any group recording deflection for a stationary conductor has a wiring error or a misread; the teacher re-checks their connections immediately. A second indicator: at least 80% of groups should record a larger deflection for fast movement than slow movement in the repeated scenario (c) trials — this sets up Lesson 5.

	galvanometer. Then move it downward — read galvanometer. (d) Move conductor at approximately 45° to the field — read galvanometer. For each step learners record in a results table: Scenario | Galvanometer Reading (L-deflect / Centre / R-deflect) | Estimated Magnitude (large / small / none) | Observation notes. After completing all four steps, each group repeats scenario (c) at FAST speed and at SLOW speed, recording whether magnitude changes. Role rotation: Conductor Handler → Galvanometer Reader → Recorder → Timer/Reporter.	step (c), ask: 'Is your conductor actually cutting field lines, or moving along them? Show me the direction again.' After 20 minutes, give a 2-minute warning: 'Finish your last reading and complete your table.' Then call the class to attention: 'Recorders, bring your results tables to the front — we are going to build a class data wall.' Stick four large paper columns on the board (one per scenario). Cold-call each group's Reporter to announce their deflection result. Tally results in each column. Ask the class: 'Look at the board. Where do ALL groups agree — deflection or no deflection?' WAIT 12 seconds. Cold-call three learners: 'Kamau — what pattern do you see? ... Fatuma — do you agree? ... Njoroge — can you say it in one sentence?'		
Phase 3 — Explain 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually complete a CER (Claim–Evidence–Reasoning) frame on a structured worksheet: CLAIM: 'A current is induced in a conductor when…' EVIDENCE: (learners select 2–3 specific results from their table, e.g. 'In scenario (a), the galvanometer read zero. In scenario (c), it deflected to the right.') REASONING: 'This shows that… because…' After 6 minutes of individual writing, learners share their CER with their group and agree on one GROUP CER to write on a mini-whiteboard. Groups display boards. Teacher reads each board aloud, highlighting commonalities. Class agrees on a SHARED STATEMENT written on the board:	Distribute CER worksheets. Say: 'You have 6 minutes — individual work only. Use your results table as your evidence. No copying from neighbours.' After 6 minutes: 'Now share within your group. You have 4 minutes to write ONE group CER on your mini-whiteboard — it must include at least one specific number or direction from your results table.' After groups display boards, read each aloud and annotate similarities in green marker. Say: 'Notice that almost every group used the word MOVING or CUTTING. Why is that word so important?' WAIT 12 seconds. Cold-call 4 learners: 'Achieng — why is the word cutting important? ... Maina — what would happen if the conductor only moved alongside the field lines, not across them? ... Zipporah — connect this to the Kericho	Claim–Evidence–Reasoning (CER) Frame — scaffolds learners from raw observations to a causal explanation. The explicit phenomenon connection step prevents the explanation from remaining abstract, anchoring the science concept to the Kericho context and satisfying the NGSS Science and Engineering Practice of Constructing Explanations.	Teacher reads the PHENOMENON CONNECTION sentences as learners write — the observable indicator is whether learners use causal language ('because', 'when…then', 'this means') rather than just descriptive language ('the light comes on'). Learners who write only description are flagged for a targeted follow-up question: 'Why does the light come on only when the wheel moves?' — pushing toward a mechanistic explanation.

	'An e.m.f (and therefore a current) is induced in a conductor only when the conductor is MOVING and CUTTING across magnetic field lines. When the conductor is stationary or moving parallel to field lines, no e.m.f is induced.' Teacher then makes the phenomenon connection explicit: 'Go back to Akoth in Kericho. She removed the dynamo from her bicycle. When she spins the wheel — what is happening inside the dynamo?' WAIT 12 seconds. 'And when she holds it still — what happens to the conductor's relationship with the field?' Learners write a 3-sentence PHENOMENON CONNECTION in their notebooks linking the experiment result to (i) the bicycle dynamo LED, and (ii) the Kericho school generator.	generator. What is the conductor doing inside that generator? ... Oduya — and what happens to the electricity supply when the generator stops spinning?' Write the SHARED STATEMENT on the board slowly, pausing at key words for learners to call them out: 'An e.m.f is induced only when the conductor is…?' (MOVING) '…and it must be…?' (CUTTING field lines). Then assign the 3-sentence PHENOMENON CONNECTION: 'Write this in your own words — 3 sentences, 3 minutes, notebooks open.'		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes — one GREEN ('I can now explain…') and one ORANGE ('I still wonder…'). On the GREEN note, learners write one thing from today's experiment that they can now explain about the Kericho phenomenon. On the ORANGE note, learners write one remaining question — e.g. 'Does the speed of spinning change the amount of electricity?', 'Why does moving parallel to the field produce no current?', 'How does Kenya Power control the speed of their generators?'. Learners post their notes on the class DQB under the driving question banner. Teacher reads 4–5 aloud, groups them into clusters ('Speed / Magnitude questions', 'Direction questions', 'Real-world application questions') and previews: 'Your orange questions about speed and	Distribute sticky notes. Say: 'GREEN note — one thing you can NOW explain about Akoth's dynamo that you could NOT explain at the start of today. ORANGE note — one question that today's experiment has raised in your mind. You have 3 minutes.' After posting, read 4–5 notes aloud — choose at least one GREEN and at least two contrasting ORANGEs. Cluster ORANGEs on the DQB with a marker: 'These three questions are all about SPEED and HOW MUCH electricity — we will answer them in Lesson 5. These two questions are about DIRECTION of current — we will come back to those in Lesson 6.' Say: 'Notice that our DQB is growing. At the start of this unit, Akoth could not explain anything. Look at how many green notes we now have. Look at how	Driving Question Board Revision — making thinking visible tracks the cumulative growth of the class's explanatory model across the 8-lesson sequence. Clustering ORANGE notes by theme previews the lesson sequence and gives learners agency in the scientific storyline, supporting the NGSS Practice of Asking Questions and Defining Problems.	Teacher scans GREEN notes for quality of explanation — the observable indicator is specificity: 'The dynamo light goes out because the conductor stops cutting field lines so no e.m.f is induced' is a strong green note; 'I understand electricity' is not. Teacher photographs the DQB at end of lesson to track which questions have been addressed across the sequence.

	magnitude — that is exactly what Lesson 5 will answer.'	many new questions we have discovered — that is what scientists do. Every answer opens new questions.'		
Phase 5 — Model Building 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative PHENOMENON MODEL — a diagram they have been building since Lesson 1, showing the Kericho generator/dynamo system. Today they add a new annotated layer: they draw the conductor between the magnet poles in position (c) (moving upward) and add arrows showing: (1) the direction of conductor movement, (2) the magnetic field lines between poles (N→S), (3) a curved arrow on the galvanometer showing deflection, and (4) a label reading 'INDUCED E.M.F — caused by conductor cutting field lines'. They cross out or revise any part of their earlier model that now conflicts with today's evidence (e.g., some learners may have previously drawn a battery as the source of current in the dynamo — this must now be replaced with the label 'relative motion = source of e.m.f'). Learners add a model legend entry: 'Condition for induction: conductor must MOVE and CUT field lines.' Finally, learners write one sentence at the bottom of their model: 'What my model still cannot explain: ___' — preparing a bridge to Lesson 5.	Say: 'Take out your phenomenon models. You have 8 minutes to add today's findings as a new layer. You must: (1) draw the conductor in scenario (c), (2) label the induced e.m.f, (3) correct anything in your earlier model that today's experiment proved wrong, and (4) write what your model still cannot explain.' Circulate and check that learners are (a) annotating position (c) — not (a) or (b) — since that is the key induction scenario, and (b) explicitly removing or revising any earlier incorrect labels. After 8 minutes, cold-call two learners to hold up their models and describe one addition: 'Wambui — what did you add today and what did you correct? ... Abubakar — what does your model still NOT explain?' Say: 'In Lesson 5, we will discover how changing the speed, the number of turns, and the strength of the magnet changes the SIZE of the induced e.m.f — and we will be one step closer to explaining how Kenya Power produces 240 volts at your home socket.'	Cumulative Annotated Model Revision — iterative model building across lessons is the primary tool for tracking conceptual growth in this sequence. Requiring learners to CORRECT previous model errors (not just add new information) develops the scientific practice of Developing and Using Models and reinforces that models are provisional and evidence-responsive.	Teacher checks that model revisions include the label 'induced e.m.f' AND a movement arrow — both must be present to indicate that the learner understands induction as a process requiring motion, not just a static field. Learners whose models still show a battery or chemical source for the dynamo current need individual follow-up before Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and note responses in your professional journal:\n1. PREDICTION ACCURACY: What fraction of learners correctly predicted zero deflection for a stationary conductor (scenario a)? If fewer than 50% predicted this correctly, learners may still hold the misconception that a magnetic field alone — without motion — produces current. Consider opening Lesson 5 with a brief revisit of this scenario before moving to quantitative factors.\n2. EXPERIMENT QUALITY: Did all groups successfully obtain a measurable deflection in scenario (c)? If any group could not, identify whether the issue was apparatus (galvanometer sensitivity, loose connections) or procedural (conductor not genuinely cutting field lines). Record this for lab technician communication before Lesson 5.\n3. CER

QUALITY: Did learners' written CLAIM statements include the word 'moving' or 'cutting' — or did they write only 'when you connect the wire'? The absence of motion language in the claim indicates the core idea has not transferred. These learners need a targeted one-on-one or small-group conversation before Lesson 5.\n4. PHENOMENON CONNECTION: Were learners able to map the experiment result onto BOTH the bicycle dynamo AND the Kericho school generator? If learners could explain one but not the other, they may be pattern-matching rather than applying a transferable principle.\n5. DQB ORANGE NOTES: Count the number of orange notes asking about speed or magnitude vs. direction vs. real-world application. This ratio tells you whether learners are ready for Lesson 5 (quantitative factors) or whether more qualitative consolidation is needed.\n6. GENDER AND PARTICIPATION: Review your cold-call log. Did you achieve gender balance across the five phases? In mixed groups, did girls take the Conductor Handler role (an active, high-status role) as frequently as boys? If not, assign roles explicitly in Lesson 5.\n7. KENYAN CONTEXT INTEGRATION: Did learners spontaneously reference the Kericho generator, Kenya Power, or bicycle dynamos during the explain phase — or did the phenomenon connection feel forced? Authentic integration is the goal; if learners needed prompting, consider opening Lesson 5 with a short real image or 60-second audio clip of a Kenyan power station turbine room.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that a straight copper conductor held stationary between the poles of a U-shaped magnet produces zero galvanometer deflection; that moving the conductor parallel to field lines produces zero or negligible deflection; that moving the conductor vertically upward (cutting across field lines) produces a clear deflection; and that moving it at an angle produces a deflection of intermediate magnitude. Learners also observed that faster movement in scenario (c) produces a larger deflection than slower movement.
What did I learn?	Learners learned that an electromotive force (e.m.f) is induced in a conductor only when there is relative motion between the conductor and the magnetic field, and specifically only when the conductor cuts across field lines rather than moving parallel to them. This is the principle of electromagnetic induction. The direction of deflection reverses when the conductor moves downward instead of upward, indicating that the direction of induced current depends on the direction of motion. Learners connected this principle to the Kericho school generator and the bicycle dynamo: in both cases, continuous rotational motion keeps the conductor cutting field lines, sustaining a continuous induced e.m.f and therefore a continuous current.
How does this explain the phenomenon?	Learners can now explain why Akoth's bicycle dynamo LED lights up only when the wheel is spinning: rotation causes the conductor inside the dynamo to continuously cut through magnetic field lines, inducing an e.m.f that drives current through the LED. When the wheel stops, the conductor becomes stationary relative to the field, the cutting action ceases, the induced e.m.f drops to zero, and the LED goes out. The same principle applies to the Kericho school generator: the large copper coils must rotate continuously relative to the fixed magnets (or electromagnets) to sustain the induced e.m.f that powers the dormitory lights, water pumps, and kitchen equipment. No battery or chemical reaction is needed — mechanical motion is converted directly into electrical energy through electromagnetic induction.

LESSON 5 (80 minutes (2 × 40-minute lessons)): Factors Affecting the Magnitude of Induced E.M.F.	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to investigate, quantify and explain the factors that affect the magnitude of induced electromotive force (e.m.f.) in an electromagnetic induction system.
Knowledge	Learners will be able to: (i) state the four factors that affect the magnitude of induced e.m.f. — speed of conductor movement, strength of the magnet, number of turns of wire, and angle of movement through the field; (ii) describe qualitatively how each factor increases or decreases the induced e.m.f.; (iii) connect Faraday's Law to real-world generators, including Kenya's national-grid power stations and bicycle dynamos.
Skills	Learners will be able to: (i) design and conduct a guided controlled investigation, changing one variable at a time while holding others constant; (ii) record galvanometer deflections systematically in a results table; (iii) plot and interpret line graphs of induced e.m.f. versus each factor; (iv) draw evidence-based conclusions and compare them with earlier predictions on the DQB.
Attitudes	Learners will: (i) appreciate that electricity generation is a product of deliberate, investigable physical principles — not magic; (ii) show curiosity and persistence when results do not match predictions; (iii) value accurate, honest data recording; (iv) recognise the socio-economic importance of understanding electromagnetic induction for Kenya's energy future (rural electrification, solar-wind hybrid farms in Turkana, geothermal at Olkaria).
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lessons 1–4 learners observed electromagnetic induction, defined induced e.m.f., and built basic models. They noticed that the student in Kericho saw the light go out when the dynamo stopped. Now in Lesson 5 they go further: they systematically vary the conditions of induction to discover WHAT controls HOW MUCH electricity is produced — directly explaining why faster cranking of the dynamo torch makes the LED brighter, and why power-station engineers choose powerful magnets and thousands of coil turns.
Safety Notes	1. Use only low-voltage demonstration galvanometers; no mains electricity during learner activities. 2. Handle bar magnets carefully — keep away from mobile phones, data-storage devices and watches. 3. Secure all connecting wires with crocodile clips to avoid short circuits. 4. If using a bicycle dynamo, ensure the wheel is clamped securely before spinning. 5. Learners with pacemakers should not handle strong magnets — alert the teacher in advance.

B. LESSON OVERVIEW

Learners conduct four sequential, teacher-guided investigations on a single experimental rig (galvanometer + coil + magnet) to discover how speed of conductor movement, magnet strength, number of coil turns, and angle of movement each affect the magnitude of induced e.m.f. Results are recorded in structured tables, plotted as bar/line graphs, and interpreted using Faraday's Law. All findings are explicitly connected back to the Kericho boarding-school phenomenon: the student's dynamo torch and Kenya's national-grid generators. The DQB is updated — earlier guesses are replaced with evidence-based statements. Learners close by revising their cumulative model of electromagnetic induction to include magnitude-controlling factors.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict 📹 **VIDEO:** [Induced current in a wire](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners individually read a short scenario card: 'Amina in Kericho has three bicycle dynamos. Dynamo A has a weak magnet and 50 turns of wire. Dynamo B has a strong magnet and 50 turns. Dynamo C has a strong magnet and 200 turns. She spins each at the same speed. Which lights the LED most brightly? Which goes out first when she slows down?' Learners write a rank order (A, B, C) and one sentence of reasoning in their science journals. They then share with a neighbour (Think-Pair). The teacher collects a quick show-of-hands class poll — tally on the board: 'How many predict C is brightest?' This visible spread of predictions creates cognitive dissonance and motivates the investigation.	1. Distribute scenario cards face-down. Say: 'Do NOT flip the card until I say go.' (Builds anticipation.) 2. Say: 'Flip. Read silently. Write your rank order and one reason. You have 90 seconds — go.' (Sets time pressure; circulate and read over shoulders without commenting.) 3. After 90 seconds: 'Turn to your partner. 30 seconds each — share your rank and reason. Disagree respectfully.' (WAIT 10 seconds for noise to settle.) 4. Poll: 'Hands up — who has C brightest?' Tally. 'Who has A brightest?' Tally. Record on board: PREDICTION TALLY. Say: 'We now have a claim. Science needs evidence. Let us find out.' 5. Cold-call 2 learners who gave DIFFERENT predictions: 'Zawadi — tell us your reasoning in one sentence.' [WAIT 12 seconds.] 'Kipchoge — do you agree or disagree?' [WAIT 10 seconds.] 6. Point to the DQB: 'Our question from Lesson 2 was: What controls how much electricity is produced? Today we answer it with data.'	PREDICT-THEN-INVESTIGATE (aligned with NGSS SEP 1 — Asking Questions and SEP 3 — Planning Investigations). By committing a written prediction before observing, learners activate prior knowledge from Lessons 1–4 and create a personal stake in the outcome. The class tally makes disagreement visible and public, motivating careful observation.	Observable indicator: Every learner has a written rank order AND a reasoning sentence in their journal before the poll. Teacher scans journals during circulation. Red flag: learners who write 'I don't know' — seat them next to a confident predictor for the investigation phase. Note the dominant wrong prediction (usually A or B) to revisit during Explain phase.
Phase 2 — Observe (Guided Investigation) 📹 **VIDEO:** [Induced current in a wire](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic](#)	Groups of 4 rotate through four mini-investigations at a shared rig (galvanometer, multi-turn coil mounted on a retort stand, set of bar magnets, connecting wires). Each investigation changes ONE variable while holding the others constant. INVESTIGATION 1 — Speed: Learners push a bar magnet into the coil slowly, then quickly; they read and record the galvanometer needle deflection angle (small/medium/large). INVESTIGATION 2 — Magnet Strength: Using one magnet vs. two stacked magnets (same	1. Demonstrate Investigation 1 at the front: 'Watch the needle — I push SLOWLY.' [Pause 5 seconds.] 'Now FAST.' Ask: 'What changed?' [WAIT 12 seconds. Cold-call 1 learner.] 2. Distribute pre-printed results tables and investigation guide cards. Say: 'You have 25 minutes. Rule: only ONE variable changes per investigation. If you change two things at once, your evidence is worthless — like testing two medicines at the same time on a patient.' (Kenyan clinical context resonates.) 3. Circulate every 4	STRUCTURED INQUIRY WITH CONTROLLED VARIABLES (NGSS SEP 3 — Planning and Carrying Out Investigations; SEP 4 — Analysing and Interpreting Data). Using a pre-printed table forces disciplined variable control. The bar chart transforms numbers into a visual pattern, making the trend immediately apparent even for learners who struggle with abstract reasoning. The side-by-side dynamo/generator photos keep the phenomenon anchoring every observation.	Observable indicators: (a) Results table has exactly one variable changing per investigation — any group with two changing variables is flagged for re-do. (b) Bar chart is plotted with labelled axes and at least 8 of 10 bars correctly placed. (c) When teacher probes 'What are you holding constant?' learner can name at least two fixed variables. Red flag: groups that push the magnet at different depths — prompt them to mark a tape line on the magnet showing 'how far to insert'.

Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	speed, same coil); record deflection. INVESTIGATION 3 — Number of Turns: Use 20-turn coil vs. 50-turn vs. 100-turn coil (same magnet, same speed); record deflection. INVESTIGATION 4 — Angle of Entry: Push the magnet straight in (90°), at 45°, and parallel to the coil axis (0°); record deflection. All results enter a pre-printed table. Learners plot a grouped bar chart on grid paper: factor on x-axis, deflection (proxy for e.m.f.) on y-axis. Groups annotate each bar: 'higher e.m.f.' or 'lower e.m.f.' Connection to phenomenon: teacher posts a photo of the Kericho generator coils on the board — 'That machine in the generator room is doing exactly what your coil is doing right now.'	minutes. Probe groups with: 'Which variable are you currently changing?' 'What are you holding constant?' 'What does a larger needle deflection tell us about the e.m.f.?' [WAIT 10 seconds each.] 4. If a group records inconsistent results: 'Repeat that trial — scientists in Kenya's GDC geothermal plant repeat measurements three times before trusting data.' 5. At 20 minutes, call a 2-minute pause: 'Pencils down. Look at your table so far. Which factor caused the BIGGEST change in deflection? Whisper to your partner.' [WAIT 10 seconds.] Cold-call 2 groups. 6. At 25 minutes: 'Transfer your data to the bar chart. Label every bar. You have 5 minutes.' 7. Post the dynamo-torch photo beside the generator photo: 'Same principle — different scale.'		
Phase 3 — Explain 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group presents their bar chart to the class (60-second gallery-walk format — charts posted on the wall; learners walk and add a sticky note 'agree' or 'question' to each chart). Teacher then leads a whole-class Claim-Evidence-Reasoning (CER) discussion: groups co-construct four scientific statements — one per factor — using the sentence frame: 'When [factor] increases, the induced e.m.f. [increases/decreases] because [reason linked to rate of flux change].' Teacher introduces Faraday's Law in plain language: 'The faster the magnetic field through the coil changes, the larger the e.m.f. — every factor we tested today was really just a way to change the RATE of flux change.' Learners then apply	1. Set gallery-walk timer: '60 seconds at each chart — one sticky note per chart.' (Keeps pace brisk.) 2. Reconvene. Say: 'Let us build our four science statements together.' Write the CER frame on the board. Call on groups sequentially — cold-call 4 different learners (one per factor): 'Nekesa — give me Statement 1 for speed.' [WAIT 12 seconds.] Scribe their words exactly, then refine together. 3. Introduce Faraday: 'All four factors share one idea — rate of change of magnetic flux. Let me show you why with a quick sketch.' Draw coil + field lines; show how more turns = more lines cut per second. 4. Ask: 'So why does the Kericho generator have THOUSANDS of turns and huge electromagnets?' [WAIT 15 seconds. Cold-call 2 learners.] 5.	CLAIM-EVIDENCE-REASONING (CER) FRAMEWORK (NGSS SEP 6 — Constructing Explanations; SEP 7 — Engaging in Argument from Evidence). The CER frame scaffolds learners from raw data to scientific explanation. Gallery-walk adds a peer-review layer without the anxiety of formal presentations. Connecting each factor back to 'rate of flux change' builds a unifying conceptual bridge to Faraday's Law, preparing learners for quantitative treatment in later grades.	Observable indicators: (a) Each learner's journal contains a CER-formatted statement for at least 3 of the 4 factors. (b) The revised Amina rank order is C > B > A with at least one piece of evidence cited. (c) During cold-call, learner can spontaneously use the word 'rate' or 'faster change' in their explanation. Red flag: learners who write 'more turns = more electricity' without a causal mechanism — prompt: 'WHY does more turns produce more e.m.f.? What is each turn doing?'

	findings directly to the phenomenon: 'Amina's Dynamo C (strong magnet, 200 turns) is brightest because it maximises TWO factors simultaneously. The Kericho generator uses massive powerful electromagnets and thousands of coil turns — now we know WHY.' Learners update Amina scenario: they write the correct rank order with a full CER justification.	Pose the revision task: 'In your journal — rewrite Amina's rank order. This time use evidence from our investigation. Use the CER frame. 3 minutes, silent.' Circulate and read. 6. Select one journal response to read aloud (with learner's permission): 'Akinyi's answer is a strong scientific claim — listen to how she uses evidence.'		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Induced current in a wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the physical DQB (large paper on the classroom wall, started in Lesson 1). Learners identify questions from earlier lessons that today's evidence can now answer. Each group selects one sticky note (written in an earlier lesson) and replaces it — or annotates it in a different colour — with an evidence-based answer. Examples of questions now answerable: 'Does spinning faster make more electricity?' → 'YES — Investigation 1 showed higher galvanometer deflection at faster speed, meaning higher induced e.m.f.' 'Why does a bigger generator make more electricity?' → 'More turns of wire + stronger magnets = larger e.m.f. (Investigations 2 and 3).' Learners also ADD two new questions generated today: e.g., 'Is there a mathematical formula for e.m.f.?' and 'What happens if we spin the coil instead of moving the magnet?' These carry the DQB forward into Lessons 6–8.	1. Direct learners to the DQB: 'Scan the board. Find one question your group's evidence from today can answer. You have 90 seconds.' 2. Groups send one representative to annotate the board simultaneously (saves time). 3. Teacher reads 2–3 updated notes aloud: 'Listen to what Group Rift Valley wrote — this is evidence-based science.' 4. Ask the class: 'What NEW questions did today's investigation raise for you?' [WAIT 15 seconds. Cold-call 3 different learners — target learners who have not spoken yet.] Scribe new questions on fresh sticky notes in a new colour. 5. Point to the driving question at the top of the DQB: 'How does moving a wire through a magnetic field produce electricity — and how is this powering homes, farms and industries across Kenya?' Ask: 'Are we closer to answering this? What do we still not know?' [WAIT 12 seconds.] 6. Summarise: 'Today we moved from OBSERVATION to EVIDENCE. Next lesson we explore HOW this principle is engineered into real generators — Olkaria, Tana River, and the Kipevu power station in Mombasa.'	DRIVING QUESTION BOARD REVISION (anchored in phenomena-based learning). Replacing prediction-sticky-notes with evidence-based answers makes conceptual growth visible and motivating. Adding new questions models authentic scientific thinking — investigation raises as many questions as it answers. The connection to Kenyan power stations (Olkaria geothermal, Tana River hydroelectric, Kipevu diesel) grounds abstract physics in national economic relevance.	Observable indicators: (a) Every group replaces or annotates at least one earlier sticky note with a statement containing the words 'because' or 'evidence showed'. (b) At least 2 new learner-generated questions are added to the DQB. (c) When teacher asks 'Are we closer to answering the driving question?' at least 3 learners can articulate a specific piece of evidence from today. Red flag: groups that simply remove a question note without writing an answer — prompt: 'What is your evidence? Write it on the note before you post it.'

Phase 5 — Model Building 📹 **VIDEO:** [Induced current in a wire] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative electromagnetic induction model started in Lesson 2 (a drawn diagram showing magnet, coil, galvanometer and arrows for induced current). They now REVISE it to add four labelled annotations showing each factor's effect on e.m.f. magnitude: (1) a double-headed arrow labelled 'faster movement → larger e.m.f.'; (2) a larger magnet symbol labelled 'stronger magnet → larger e.m.f.'; (3) a coil with more turns labelled 'more turns → larger e.m.f.'; (4) an angle indicator labelled 'perpendicular movement (90°) → maximum e.m.f.; parallel movement (0°) → zero e.m.f.' Learners also annotate their model with a Kenyan engineering note: 'The Olkaria IV geothermal generator uses steam to spin coils with thousands of turns inside powerful electromagnets — all four factors are maximised.' Finally, learners write a 2-sentence 'model statement' below the diagram: 'My model now shows… Before Lesson 5 my model was missing…'	1. Say: 'Take out your model from Lesson 2. Look at it carefully. What is missing that we now know?' [WAIT 12 seconds. Cold-call 1 learner: 'Ochieng — what will you add first?'] 2. Give instruction: 'Add all four factor annotations in a DIFFERENT colour so the revision is visible. You have 8 minutes.' Circulate — check that each annotation includes a direction of effect ('larger' or 'smaller'), not just a label. 3. Prompt deeper annotation: 'Can you show on the diagram WHERE the angle of entry matters? Draw the magnet at 90° and at 0° and compare the arrow sizes for e.m.f.' 4. At 6 minutes: 'Now write your 2-sentence model statement below the diagram. Start with: My model now shows…' 5. Pair-share model statements: 'Read your statement to your partner. Partner — give one piece of feedback: one thing that is strong and one thing to add.' [WAIT 10 seconds for feedback exchange.] 6. Close with the phenomenon: 'The student in Kericho could not explain why faster spinning made the LED brighter. Look at your model. Can YOU explain it now? Turn to a partner and explain it in 20 seconds — go.' [WAIT 20 seconds.] Cold-call 1 pair to share their explanation with the class.	CUMULATIVE MODEL REVISION (NGSS SEP 2 — Developing and Using Models). Using a distinct colour for revisions makes the learning journey visible — learners literally see their understanding grow. The 'model statement' (My model now shows… Before Lesson 5 my model was missing…) is a metacognitive tool that consolidates understanding and prepares for the summative model review in Lesson 8. The 20-second partner explanation is a low-stakes retrieval practice that strengthens memory consolidation.	Observable indicators: (a) Revised model shows all four factor annotations in a different colour from the original. (b) Each annotation includes a directional effect (e.g., 'increases e.m.f.' not just 'speed'). (c) The angle annotation correctly shows 90° → maximum and 0° → zero e.m.f. (d) The 2-sentence model statement explicitly names at least 2 factors. Red flag: learners whose models are identical to their Lesson 2 version — require them to stay for a 5-minute guided revision session immediately after class.

D. TEACHER REFLECTION

After the lesson, ask yourself: 1. PREDICTION SPREAD: Was there genuine disagreement in the class prediction poll? If all learners predicted correctly before investigating, the scenario card may be too easy — revise for a future cohort to include a counter-intuitive twist (e.g., a faster-moving weak magnet vs. a slow-moving strong magnet). 2. INVESTIGATION QUALITY: Did all groups successfully hold variables constant in all four investigations? If groups with Investigation 4 (angle) struggled, consider pre-marking the magnet with coloured tape at 0°, 45° and 90° entry positions before the next delivery. 3. CER STATEMENTS: Did learners use causal language ('because the rate of flux change increases') or only correlational language ('when speed goes up, e.m.f. goes up')? If only correlational — revisit the Faraday explanation at the start of Lesson 6 with a quick 3-minute recap. 4. DQB VITALITY: Were the new learner-generated questions scientifically productive (pointing

toward Lessons 6–8 content) or too vague? If vague, use a 'question starter' scaffold next time: 'I wonder whether…' or 'What would happen if…' 5. KENYAN CONTEXT ENGAGEMENT: Did the Olkaria/Tana River references spark curiosity or confusion? If learners were unfamiliar with these plants, assign a 5-minute reading card on Kenya's electricity generation mix as a homework bridge to Lesson 6. 6. EQUITY CHECK: Which learners dominated the galvanometer readings? Ensure group roles (reader, recorder, variable-controller, timekeeper) rotate so all learners handle apparatus. Note any gender imbalance for redistribution in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed galvanometer needle deflections change in magnitude as they varied: (a) the speed of pushing a bar magnet into a coil — faster push → larger deflection; (b) the number of stacked magnets — two magnets → larger deflection than one; (c) the number of coil turns — 100-turn coil → larger deflection than 20-turn coil; (d) the angle of magnet entry — 90° (perpendicular) → maximum deflection; 0° (parallel) → zero deflection. Learners also observed that the galvanometer needle returned to zero immediately when movement stopped — reinforcing the Kericho dynamo observation that the LED goes out when the wheel stops.
What did I learn?	Learners learned that four factors control the magnitude of induced e.m.f.: speed of conductor/magnet movement, strength of the magnetic field, number of turns of wire in the coil, and angle of movement through the field. They learned that all four factors share a single underlying principle — Faraday's Law — that the induced e.m.f. is proportional to the rate of change of magnetic flux through the coil. Learners also learned how to design a fair investigation by controlling variables, how to record results systematically in a table, how to plot and read a bar chart of electromagnetic data, and how to construct a Claim-Evidence-Reasoning explanation from experimental results.
How does this explain the phenomenon?	Learners explained that the student in Kericho sees the LED go brighter when she cranks the dynamo faster because increasing speed increases the rate at which magnetic field lines are cut by the coil, increasing the induced e.m.f. They explained that Kenya's large power-station generators (e.g., Olkaria IV geothermal, Tana River hydroelectric) produce thousands of volts because engineers maximise all four factors simultaneously: rotating coils at high speed, using powerful electromagnets, winding thousands of turns of copper wire, and designing the geometry so conductors always cut field lines at 90°. Learners used the CER framework to replace their earlier predictions on the DQB with evidence-based statements, and revised their cumulative electromagnetic induction model to show all four magnitude-controlling factors with directional annotations.

LESSON 6 (80 minutes (2 × 40-minute lessons)): Applications of Electromagnetic Induction in Kenya	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to connect experimental principles of electromagnetic induction to real-world applications that power Kenyan homes, farms and industries — including AC generators, transformers, the Olkaria Geothermal Plant, the Lake Turkana Wind Farm, electric bells and bicycle dynamos.
Knowledge	Learners will be able to: (a) describe how AC generators, transformers, the Olkaria Geothermal Plant turbines, Lake Turkana Wind Farm turbines, electric bells and bicycle dynamos all rely on electromagnetic induction; (b) explain how Kenya Power's national grid transmits electricity using transformers based on the principle of electromagnetic induction; (c) link each application to the experimental factors (speed of motion, number of coil turns, magnetic field strength) established in Lessons 4 and 5.
Skills	Learners will be able to: (a) use digital devices and print media to research real-life applications of electromagnetic induction; (b) prepare and deliver a short group presentation linking an application to experimental evidence; (c) construct an annotated diagram or model showing how an AC generator or transformer operates; (d) evaluate how changes in generator design variables would affect electrical output in the Kenyan context.
Attitudes	Learners will: (a) appreciate how electromagnetic induction underpins Kenya's national electricity infrastructure and everyday devices; (b) show curiosity and patriotism when investigating how Kenyan energy installations harness the same physics demonstrated in class; (c) demonstrate responsibility and respect by contributing equitably to group research and presentations.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lessons 4 and 5 the student from Kericho proved experimentally that moving a conductor in a magnetic field induces an e.m.f. and identified the factors that control its magnitude. In Lesson 6 she now asks: where exactly is this principle at work in the real world around her? By researching and presenting on AC generators, transformers, Olkaria, Lake Turkana, electric bells and bicycle dynamos, learners build the conceptual bridge between the school laboratory and Kenya's national grid — directly answering the driving question at scale.
Safety Notes	No high-voltage equipment is handled in this lesson. If learners bring in a bicycle dynamo for demonstration, ensure the wheel axle is secured before spinning. Supervise handling of any magnets near electronic devices (tablets, phones). Remind learners to cite sources responsibly when using the internet and to observe cyber-safety guidelines.

B. LESSON OVERVIEW

Lesson 6 is a research-and-presentation lesson that bridges laboratory evidence (Lessons 4–5) to real-world applications of electromagnetic induction in Kenya. Working in six expert groups, learners use digital devices and print media to investigate one assigned application each: (1) AC generators, (2) step-up and step-down transformers, (3) Olkaria Geothermal Plant, (4) Lake Turkana Wind Farm, (5) electric bells, and (6) bicycle dynamos. Each group prepares a 4-minute presentation linking their application explicitly to the experimental factors established in previous lessons. A gallery-walk debrief consolidates understanding across all six applications. The lesson closes with learners updating the class Driving Question Board and revising their cumulative explanatory models to show how Kenya Power's national grid depends entirely on electromagnetic induction. The anchoring phenomenon — the Kericho student watching copper coils spin inside the generator room — is revisited throughout every phase.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write a 'prediction card': they list at least THREE places in Kenya (beyond the school generator and their bicycle dynamo) where they think electromagnetic induction is happening right now. They then rank their three choices from 'most sure' to 'least sure' and briefly explain their reasoning in one sentence per choice. Pairs share their cards and identify any applications they both listed.	Display the anchoring phenomenon image (Kericho generator room / bicycle dynamo LED) on the board. Say: 'Our Kericho student has now proven in the lab that moving a coil in a magnetic field induces an e.m.f. — and that speed, coil turns and field strength all matter. Today we ask: where in Kenya is this happening RIGHT NOW, at much larger scale?' Distribute prediction cards. Give WAIT TIME of 12 seconds after posing the prompt: 'Think silently — name three real places or devices in Kenya that you believe use electromagnetic induction today.' After individual writing (3 min), say: 'Turn to your partner. Compare your lists. Do you agree on any? Disagree on any?' (2 min pair share). Cold-call 4 pairs: 'Wanjiku and Ochieng, what was on both your lists?' Record all suggested applications on the whiteboard under the heading 'Our Predictions'. Do not correct or affirm — say: 'We will test each of these claims today using evidence.'	Predict-then-Investigate (Elicitation of Prior Knowledge): Making predictions activates what learners already know from daily life — matatu charging systems, Kenya Power pylons, phone chargers — and creates a personal stake in the investigation. Recording predictions publicly on the board makes thinking visible and gives the class a shared reference point to revisit during the debrief.	Teacher circulates during individual writing and reads prediction cards over shoulders. Observable indicator: Can each learner name at least one application beyond the bicycle dynamo? Note any learner who writes only 'bicycle' or leaves the card blank — these learners are paired with stronger peers during the group research phase. Collect a sample of 6 prediction cards to review after the lesson.
Phase 2 — Observe (Research and Evidence Gathering) 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners are organised into six expert groups (4–5 learners each). Each group is assigned one application card: Group 1 — AC Generators (how mechanical rotation produces alternating current); Group 2 — Transformers (step-up at Olkaria substation, step-down into homes); Group 3 — Olkaria Geothermal Plant, Naivasha (steam drives turbines); Group 4 — Lake Turkana Wind Farm, Marsabit (wind drives	Before releasing groups, model the research guide card using the bicycle dynamo as an example for 2 minutes: 'Group 6 will answer Q2 by saying that a bicycle dynamo maximises speed — because our Lesson 5 experiment showed that faster movement of the conductor produces a larger e.m.f. This is exactly why faster pedalling makes the light brighter.' Circulate continuously during research (target: visit every group at least	Jigsaw Expert Groups with Structured Research Cards: Each group becomes the class 'expert' on one application, ensuring that all six applications are covered within one lesson without overwhelming any single group. The structured research questions force every group to explicitly link their application to the prior experimental evidence — preventing surface-level 'copy-paste' research and building	During group research: teacher checks that each group's answer to Q2 correctly names at least one factor from Lessons 4–5 (speed/rate of change of flux, number of turns, field strength). Red flag: a group that cannot answer Q2 — intervene immediately and direct them back to their Lesson 5 notes. During presentations: observable indicator — does each presenting group use the words 'relative motion',

📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	turbines, 310 MW); Group 5 — Electric Bells (electromagnet and make-and-break circuit); Group 6 — Bicycle Dynamos (permanent magnet spinning inside a coil). Each group receives: (a) a printed research guide card with 4 structured questions (see below), (b) access to a tablet or shared laptop, and (c) one or two print-media articles pre-selected by the teacher. The four structured research questions on every card are: Q1 — Describe in one paragraph how your application uses electromagnetic induction. Q2 — Which factors from our Lessons 4 and 5 experiments (speed, coil turns, field strength) does your application deliberately maximise, and why? Q3 — Draw and label a simple diagram of your application. Q4 — How does your application connect to the Kericho student's observation — i.e., how does it answer part of our driving question? Groups record findings on A3 poster paper using both text and diagrams. After 20 minutes of research, each group delivers a 4-minute presentation to the class, followed by 2 minutes of peer questions.	twice in 20 minutes). At each visit, prompt with: 'Show me where in your diagram the relative motion is happening.' and 'Which specific factor from our experiment is your application exploiting?' Use WAIT TIME of 10 seconds after each prompt before accepting responses. During presentations, stand to the side and use cold-call questions directed at the LISTENING groups: after Group 3 (Olkaria) presents, cold-call a member of Group 1: 'James, Group 3 said steam spins a turbine at Olkaria. Which factor from our experiment does spinning faster correspond to?' (Wait 12 seconds.) After all presentations, say: 'Look at the predictions board. Which of our predicted applications were correct? Which did we miss?' Update the board collaboratively.	genuine conceptual bridges. Public poster-making makes the evidence visible and reusable during the model-building phase.	'coil', 'magnetic field' and 'induced e.m.f.' at least once? Peer questioners are required to ask at least one question that references the experimental factors. Teacher records on a class checklist which groups made the experimental link explicitly and which did not.
Phase 3 — Explain (Sensemaking and Synthesis) 📹 **VIDEO:** [Introduction to experiment design] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	After all six presentations, the class participates in a teacher-facilitated whole-class discussion: 'What do ALL six applications have in common?' Learners contribute to building a class consensus statement on the whiteboard. Individually, learners then complete an 'Application Analysis Card': they choose ANY ONE of the six applications (not their own group's) and write a short paragraph (5–7 sentences) explaining how it works using the	Open the synthesis discussion by pointing to the prediction board: 'We now have evidence for six applications. What single idea connects ALL of them back to what we observed in Lessons 4 and 5?' Allow 12 seconds of silent think time, then cold-call 3 learners by name. Write key phrases from their responses on the board without editorialising. Then say: 'Let me push us further — Kenya Power's national grid transmits electricity at 132,000 volts using	Claim-Evidence-Reasoning (CER) Writing: The Application Analysis Card requires learners to make a claim (this application uses electromagnetic induction), support it with evidence (specific features of the device), and construct reasoning that chains from relative motion → changing magnetic flux → induced e.m.f. → electrical output. This moves learners from description to mechanistic explanation — the hallmark of scientific sensemaking	Collect all Application Analysis Cards at the end of the phase. Success criteria: (1) correct use of at least 4 vocabulary terms; (2) at least one sentence that explicitly links a design feature of the device to an experimental factor; (3) at least one quantitative or comparative reasoning sentence. Cards that meet all three criteria indicate the learner is approaching mastery of the application outcome. Cards missing criterion (2) or (3) identify learners who

📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	vocabulary: relative motion, magnetic flux, induced e.m.f., coil, soft iron core (where relevant), alternating current (where relevant). They must include one quantitative reasoning sentence, e.g., 'At Olkaria, the turbines rotate at higher speed so the rate of change of magnetic flux is greater, producing a larger induced e.m.f.' Pairs then swap cards and provide one piece of written peer feedback using the sentence stem: 'You correctly explained ___ but you could strengthen your explanation by ___.'	step-up transformers, then reduces it to 240 volts at your home using step-down transformers. Both the step-up and step-down transformer depend on the same principle our Kericho student observed. Can someone explain why a transformer would STOP working if the current were direct current (DC) rather than alternating current (AC)?' Give 15 seconds WAIT TIME. Cold-call 2 learners. If the concept of changing flux is not articulated, prompt: 'Think back — what condition was ALWAYS needed in every one of our Lessons 4 and 5 experiments for the galvanometer to deflect?' Guide learners to articulate: 'There must be CHANGING magnetic flux — a static field produces no induction.' During the Application Analysis Card writing (8 min), circulate and read over shoulders. Highlight strong quantitative sentences aloud: 'Listen to how Amina has connected speed to induced e.m.f. in her card — this is exactly the kind of explanation that answers our driving question.'	— and directly addresses the KICD learning outcome on appreciating the applications of electromagnetic induction in day-to-day life.	need scaffolded support in Lesson 7. Peer feedback written on the back of the card provides additional formative data on learners' ability to evaluate scientific explanations.
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** [Introduction to experiment design] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space	Learners return to the class Driving Question Board. Each group is given two sticky notes of different colours: YELLOW = 'A question this lesson has NOW answered for me' and BLUE = 'A new question this lesson has raised for me.' Groups discuss for 2 minutes, then each group nominates one member to place both sticky notes on the DQB. The class reads all new yellow and blue notes silently (2 min). The teacher facilitates a 3-minute whole-class discussion to categorise the new blue questions:	Point to the original driving question displayed at the top of the DQB: 'How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya?' Say: 'Six lessons ago, our Kericho student could not explain why her bicycle dynamo light went out when the wheel stopped. Today, what can she now explain?' Cold-call 2 learners without prior warning. After the sticky-note activity, read two or three blue notes aloud and say: 'These blue	Two-Colour Sticky-Note DQB Protocol: Using two distinct colours externalises both the consolidation of understanding (yellow) and the generation of new questions (blue) simultaneously. Categorising blue questions into physics, economics and engineering domains helps learners see that scientific phenomena have multi-disciplinary dimensions — a key CBC competency — and creates a transparent, learner-owned roadmap for the remaining lessons in the sequence.	Teacher reads all yellow sticky notes after the lesson. Observable indicators of progress: yellow notes that correctly name a specific mechanism (e.g., 'I now understand that a transformer only works with AC because DC does not change the magnetic flux') indicate conceptual gains. Yellow notes that are vague (e.g., 'I learned about electricity') identify learners needing further consolidation. Blue notes that pose mechanistic questions (e.g., 'Why does doubling coil turns double the voltage in a transformer?') indicate

by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Are they about HOW transformers work (physics detail)? About WHY Kenya uses geothermal over other sources (economics/environment)? About WHAT happens when the national grid fails (engineering)? Learners physically move blue sticky notes into these categories on the board, signalling which questions remain open for Lesson 7 and Lesson 8.	questions are exactly what good scientists ask. Which of these blue questions do you think Lesson 7 — where we look more closely at AC vs DC and transformer calculations — will answer?' Allow 10 seconds think time. Cold-call 1 learner. Highlight the blue question cluster around 'How does Kenya Power ensure constant voltage?' and say: 'This cluster tells us that understanding transformers quantitatively is our next step — and it connects directly back to the Olkaria and Lake Turkana presentations you gave today.'		readiness for quantitative treatment in Lesson 7. Blue notes about safety or environmental impact of wind farms signal strong citizenship and PCI connections.
Phase 5 — Model Building 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually retrieve the cumulative explanatory model they have been building since Lesson 4 (drawn in their science notebooks). They now ADD a second layer to their model — a 'Kenya Grid Layer' — showing how the electromagnetic induction principle, already represented in their model as a conductor moving in a magnetic field, scales up from: (a) the school lab galvanometer → (b) a bicycle dynamo LED → (c) an AC generator at Olkaria → (d) a step-up transformer → (e) the 132 kV national grid transmission line → (f) a step-down transformer → (g) the 240 V socket in the Kericho dormitory that powers the lights. Learners label each stage with the relevant experimental factor being exploited (speed, coil turns, field strength) and annotate in a different colour any stage where energy is lost (heat in transmission lines). Finally, learners write a two-sentence 'Model Claim' at the bottom of their model: 'My model shows that… because the evidence from [name a specific lesson/experiment] demonstrated	Display a blank version of the 'Kenya Grid Layer' diagram on the board (7 boxes connected by arrows, labelled a–g as above but with no annotations). Say: 'Your model is growing. In Lessons 4 and 5 you explained WHY induction happens. Today's evidence tells you WHERE it happens in Kenya. Your job now is to connect those two layers.' Give learners 8 minutes of silent individual model-building time. Circulate and use targeted prompts: 'Between stage (c) and stage (d), what specific factor is the transformer exploiting — go back to Q2 from your research card.' and 'At stage (e), 132,000 volts travels along the grid. From your Lesson 5 notes, what would happen to the induced e.m.f. if the turbine at Olkaria slowed down — and what does that mean for the entire country?' (WAIT TIME: 12 seconds before accepting responses.) With 3 minutes remaining, ask two volunteers to share their Model Claim sentences aloud. Highlight claims that use causal language ('because', 'which	Cumulative Layered Model Revision (Progressive Model Building): Adding a 'Kenya Grid Layer' to an existing model requires learners to integrate new evidence with prior knowledge rather than starting from scratch — this is the NGSS Science and Engineering Practice of Developing and Using Models. The chain from lab galvanometer to national grid makes the scaling of electromagnetic induction spatially and conceptually explicit. The two-sentence Model Claim enforces causal reasoning and provides a written artefact that can be assessed for depth of mechanistic understanding.	Teacher collects or photographs learner models at the end of the lesson. Four-level observable rubric: (4) Model correctly chains all seven stages, labels at least two experimental factors per stage, identifies an energy loss point, and the Model Claim uses causal language citing specific lesson evidence. (3) Model chains at least five stages with correct labels and a valid Model Claim. (2) Model chains three or four stages; some labels are present but the Model Claim is descriptive rather than causal. (1) Model shows only the lab apparatus from Lesson 4–5 with no extension to real-world applications. Models at levels 1–2 are flagged for targeted support at the start of Lesson 7.

	that…'	means that', 'therefore') as exemplars. Close by connecting back to the phenomenon: 'The Kericho student's question — how can spinning a coil produce enough electricity to light an entire dormitory — now has an answer that stretches all the way from the bicycle dynamo on her shelf to the Olkaria turbine hall. Your model tells that story.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. EVIDENCE LINKS: Did every group's presentation explicitly name at least one experimental factor (speed, coil turns, field strength) from Lessons 4–5? If not, which groups struggled to make this link — and what scaffolding will you provide at the start of Lesson 7?

2. KENYAN CONTEXT: Were learners genuinely engaged when the Olkaria and Lake Turkana applications came up, or did discussion remain abstract? Consider inviting a Kenya Power engineer or showing a short documentary clip of the Olkaria plant in Lesson 7 to deepen the local connection.

3. DQB QUALITY: Review the blue sticky notes. Do the new questions cluster around quantitative transformer calculations, around energy loss, or around environmental/economic dimensions? Use this data to adjust the emphasis in Lesson 7 (AC vs DC, transformer ratios).

4. MODEL PROGRESSION: Compare learners' Lesson 6 models with their Lesson 4 models. Are learners moving from purely descriptive ('a coil moves near a magnet') to mechanistic ('changing magnetic flux induces an e.m.f. proportional to the rate of change')? Learners still at the descriptive level need explicit CER scaffolding in Lesson 7.

5. GENDER AND PARTICIPATION: Did all genders participate equally in group presentations and cold-call responses? If the same 4–5 learners dominated, use a structured turn-taking protocol in Lesson 7 (e.g., talking chips or role cards).

6. PHENOMENON CONNECTION: Could every learner articulate, by the end of the lesson, how the Kericho student's generator room observation connects to the Lake Turkana Wind Farm? If not, re-open the phenomenon image at the start of Lesson 7 and explicitly narrate the chain.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners used digital devices and print media to research six real-world applications of electromagnetic induction in Kenya: AC generators, transformers, the Olkaria Geothermal Plant (Naivasha), the Lake Turkana Wind Farm (Marsabit), electric bells and bicycle dynamos. Each expert group gathered evidence and presented findings to the class, with specific reference to how each application connects to experimental data from Lessons 4 and 5.
What did I learn?	All six applications — from the bicycle dynamo on the Kericho student's shelf to the 310 MW Lake Turkana Wind Farm — operate on the same principle: relative motion between a conductor and a magnetic field induces an electromotive force. The magnitude of this e.m.f. is controlled by the same factors identified in the lab: speed of rotation, number of coil turns and strength of the magnetic field. Kenya Power's national grid depends entirely on this principle: step-up transformers at Olkaria raise voltage to 132 kV for transmission, and step-down transformers reduce it to 240 V at the point of use — both relying on changing magnetic flux in a coil.
How does this explain the phenomenon?	The Kericho student's generator room observation is no longer puzzling: the large copper coils she sees rotating inside the metal housing are the same coil-in-a-magnetic-field setup she demonstrated in the lab — just scaled up enormously. The coils spin at

	high speed (maximising rate of change of flux), have thousands of turns (maximising induced e.m.f.) and are wound around strong electromagnet cores (maximising field strength). When the wheel of her bicycle dynamo stops, relative motion ceases, the magnetic flux through the coil no longer changes, and — by Faraday's law — the induced e.m.f. immediately drops to zero, which is why the LED goes out instantly. The same physics, operating at every scale from a bicycle wheel to a geothermal turbine, is what powers Kenya's homes, farms and industries.

LESSON 7 (80 minutes (2 × 40-minute lessons)): Project: Design and Make a Simple Electric Bell	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners apply their accumulated understanding of magnetisation, demagnetisation, magnetic field patterns, electromagnetic induction, and factors affecting induced e.m.f. by designing and constructing a functional simple electric bell from locally available materials. The project consolidates all sub-strand concepts into a single artefact and makes the connection to the anchoring phenomenon explicit: the bell's electromagnet switching on and off mirrors the controlled magnetisation and demagnetisation that occurs in Kenya's large industrial generators.
Knowledge	Learners demonstrate understanding that: (i) an iron nail wrapped in copper wire and connected to a dry cell becomes a temporary electromagnet; (ii) the electromagnet attracts a metal spring strip (armature), which breaks the circuit, causing demagnetisation and the strip to spring back — this cycle produces the ringing sound; (iii) increasing the number of coil turns or current strengthens the magnetic field; (iv) the on/off switching of the electromagnet is an example of controlled magnetisation and demagnetisation in soft iron.
Skills	Learners are able to: design a labelled circuit diagram for a simple electric bell; select and safely use locally available materials (iron nail, copper wire, dry cell, metal strip, wooden base, drawing pins); construct, test and troubleshoot the bell; record the design process and test results in a structured project journal; evaluate design modifications against evidence gathered in Lessons 1–6.
Attitudes	Learners appreciate the creativity and imagination required to translate scientific principles into functional devices; demonstrate unity by sharing roles equitably within the project team; show responsibility by handling apparatus safely and managing material waste; appreciate that the same electromagnetic principle powering the bell is responsible for Kenya's national electricity supply.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	In Lessons 1–6 learners gathered evidence about magnetisation methods, magnetic field patterns, the meaning of induced e.m.f., the electromagnetic induction experiment (galvanometer, U-shaped magnet, straight conductor), and the factors affecting the magnitude of induced e.m.f. Lesson 7 is the synthesis project: learners now engineer a device — the simple electric bell — that visibly and audibly demonstrates controlled magnetisation and demagnetisation. This directly answers why the student in Kericho sees the generator coils switching states continuously, and why movement (and its cessation) controls the output. The bell's make-and-break circuit is the miniature analogue of the generator's rotating coil switching polarity with every half-turn.
Safety Notes	1. All electrical work uses low-voltage dry cells (1.5 V) only — no mains electricity. 2. Cut ends of copper wire are sharp; learners should handle with care and keep off skin and eyes. 3. Wooden bases should be sanded to avoid splinters. 4. Collect and dispose of scrap wire and insulation responsibly to avoid environmental pollution. 5. No learner should attempt to connect the circuit to a wall socket or any mains power source under any circumstances. 6. First-aid kit must be accessible in the room throughout the project session.

B. LESSON OVERVIEW

Lesson 7 is the KICD-mandated project lesson for Sub-Strand 3.4. Over two consecutive 40-minute periods, learners work in groups of four to design, build, test, troubleshoot and document a simple electric bell using locally available materials: an iron nail, copper wire (approximately 1 m, 26–28 SWG), a 1.5 V dry cell, a thin metal strip (armature — cut from a tin can lid or a paper clip straightened), a wooden base, drawing pins, and connecting wires. The lesson is structured around the five ARES phases. In the Predict phase, learners sketch how they think the bell will work before building. In the Observe phase, they construct the bell following a teacher-guided design brief and observe whether it rings. In the Explain phase, they troubleshoot failures and annotate their circuit diagrams using sub-strand vocabulary (magnetisation, demagnetisation, induced current, field strength, make-and-break circuit). In the DQB phase, they update the class Driving Question Board with new

evidence linking the bell's electromagnet to Kericho's generator. In the Model Building phase, they revise their cumulative class model to show how controlled on/off magnetisation produces both the bell sound and — at industrial scale — alternating current for Kenya's national grid. The lesson foregrounds NGSS Science and Engineering Practices: asking questions, planning and carrying out investigations, constructing explanations, and designing solutions. Core competencies developed include Creativity and Imagination, Learning to Learn, Communication and Collaboration, and Self-Efficacy. PCIs addressed include Life Skills (self-esteem) and Safety and Security.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict: How do we think the bell will work? 📹 **VIDEO:** Magnetism, light and the magneto optic Kerr effect Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner group (4 members) receives a design brief card listing available materials: iron nail (10 cm), 1 m of copper wire (insulated), 1× 1.5 V dry cell, a thin metal strip (~8 cm, cut from a tin can), a small wooden board (~15 cm × 10 cm), 4 drawing pins, and 2 connecting wires with crocodile clips. Without any additional instruction, each group spends 7 minutes sketching a labelled diagram in their project journals predicting: (a) how the components will be connected; (b) where the electromagnet will be; (c) what will cause the bell to ring repeatedly rather than just once; (d) what will happen if the wheel of the Kericho student's bicycle dynamo is held still — linking the prediction explicitly to the anchoring phenomenon. Groups then share one prediction with the class (gallery-walk style: journals open on desks, learners circulate for 2 minutes reading other groups' sketches silently).	1. Distribute design brief cards and materials kits BEFORE giving any explanation. Say verbatim: 'Before we build anything, I want each group to predict how you think this bell will work. Draw your prediction in your project journal. You have 7 minutes — go.' START a visible timer. 2. Circulate silently for the first 3 minutes, noting 2–3 groups whose diagrams show interesting or divergent predictions (one correct make-and-break idea, one that shows a continuous circuit with no break mechanism, one that omits the iron core). Do NOT correct anyone yet. 3. At 5 minutes, prompt quietly: 'Make sure your diagram addresses question (c) on the card — what makes it ring MORE THAN ONCE?' 4. After 7 minutes, call time and initiate gallery walk: 'Stand up, leave your journal open, and spend 2 minutes reading two other groups' predictions silently.' 5. After gallery walk, cold-call 3 groups (choose the divergent ones noted earlier). Ask each: 'What is the most important difference between your diagram and the one you just read?' WAIT 12 seconds after each question before accepting a response. 6. Record 3 key prediction claims on the board under the heading 'Our Predictions	Prediction-before-instruction (NGSS Practice 1 — Asking Questions and Defining Problems). By committing a design sketch to paper before building, learners activate prior knowledge from Lessons 1–6 (magnetisation, field patterns, factors affecting e.m.f.) and generate testable claims. The gallery walk surfaces the conceptual diversity in the room — particularly whether groups understand the make-and-break mechanism — giving the teacher diagnostic information without formal testing. Connecting prediction item (d) to the dynamo phenomenon maintains the storyline thread.	Observable indicator: Every group's project journal contains a labelled sketch with at least three named components and at least one written sentence addressing the make-and-break question. Teacher scans journals during circulation; any group with a blank page or only a list of materials receives the prompt: 'Show me in a drawing where the electromagnet is and what happens to the metal strip when current flows.' The three divergent predictions recorded on the board serve as a baseline against which end-of-lesson journal entries will be compared.

		— Lesson 7'. Do NOT reveal which are correct.		
Phase 2 — Observe: Build and test the simple electric bell 📹 **VIDEO:** Magnetism, light and the magneto optic Kerr effect Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups follow a structured design brief (provided on the card) to build the bell in 20 minutes. Step-by-step guidance on the card: (1) Wind the copper wire tightly around the iron nail (at least 30 turns), leaving ~10 cm free at each end — this is the electromagnet. (2) Strip insulation from the wire ends and connect one end to the positive terminal of the dry cell. (3) Fix the nail electromagnet upright on the wooden base using a drawing pin or a blob of modelling clay. (4) Bend the thin metal strip so one end is near the nail head (the armature) and the other end touches a drawing pin contact point connected to the negative terminal of the dry cell — but the strip itself is part of the circuit. (5) Adjust the gap between armature and nail so the strip is just barely NOT touching the nail when the circuit is open. (6) Close the circuit by pressing the free wire end to the dry cell terminal and observe. Learners record in their project journals: (a) what they see and hear; (b) whether the bell rings once or repeatedly; (c) any adjustments made and why. Groups that finish early attempt Modification A (add 20 more coil turns) and Modification B (increase gap between armature and nail) and record the effect on sound volume and repetition rate.	1. Before groups begin building, hold up a completed working bell (pre-built by teacher) and ring it once. Say: 'This is what a working bell sounds like. Your job is to build one that does this — and then explain WHY it works.' Do NOT show the internal wiring. 2. Release groups to build. Circulate actively. At 5 minutes, identify any group that has connected the armature permanently to the nail (no gap). Say to that group only: 'Look at your armature — is there any way it can move? What needs to happen physically to break the circuit?' WAIT 10 seconds. 3. At 10 minutes, if a group has a working bell, ask them: 'Can you explain to me in one sentence what is happening inside that nail right now?' Cold-call one member (not the group spokesperson). WAIT 12 seconds. Expect answer referencing magnetisation. 4. At 15 minutes, do a 60-second class pause: 'Freeze — hands down, tools down. Raise your hand if your bell is ringing repeatedly.' Count hands aloud. Then: 'Raise your hand if it rang once and stopped.' Count aloud. Then: 'Raise your hand if it has not rung at all.' Address the silent-bell groups with: 'Check three things — your circuit connections, the gap between the strip and the nail, and whether your wire ends are properly stripped.' 5. At 18 minutes, prompt modification groups: 'For those of you trying Modification A or B — record your observation BEFORE you change anything so you have a comparison.' 6. At 20 minutes, call	Hands-on engineering design-and-test (NGSS Practices 3 — Planning and Carrying Out Investigations, and 6 — Constructing Explanations and Designing Solutions). The build phase transforms abstract sub-strand concepts into a physical, observable artefact. The 60-second class pause at 15 minutes is a structured 'data share' that makes the distribution of outcomes visible to the whole class, turning individual group observations into collective class evidence. The pre-built teacher bell serves as an anchor for the desired outcome without scaffolding the solution path.	Observable indicators: (1) At least 3 of 5 groups have a bell producing at least one audible ring by the 15-minute check. (2) Each project journal contains at minimum: a record of the gap distance tried, a description of the sound heard, and one sentence noting whether it rang once or repeatedly. (3) During the 60-second freeze, if more than 2 groups report total silence, the teacher spends 2 minutes at the board drawing the make-and-break circuit schematic and labelling the contact point — this is the contingency scaffold.

		time. Say: 'Write two sentences in your journal: what you observed and one thing that surprised you.'		
Phase 3 — Explain: Troubleshoot, annotate and connect to the phenomenon 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Groups that have a working bell now focus on explanation. Groups that have a non-working bell spend the first 8 minutes troubleshooting with teacher support, then join the explanation phase. Explanation task (15 minutes total): Each group produces an annotated circuit diagram in their project journal that labels: (a) the electromagnet (iron nail + coil); (b) the make-and-break contact point; (c) the armature and its role; (d) where magnetisation occurs and where demagnetisation occurs; (e) a written explanation of the complete cycle: current flows → nail magnetises → armature attracted → circuit breaks → nail demagnetises → armature springs back → circuit closes again → repeat. Groups then answer two Phenomenon Connection Questions written on the board: Q1 — 'The student in Kericho holds her dynamo still and the LED goes out. Your electric bell also stops if you hold the armature against the nail permanently. What do the two situations have in common?' Q2 — 'The Kericho generator produces enough electricity for the entire dormitory. Your bell uses a 1.5 V dry cell. What changes would you make to your electromagnet to produce a stronger effect — and how does this relate to what happens inside the generator?' Groups write answers in their project journals before any whole-class discussion.	1. Write the two Phenomenon Connection Questions on the board at the START of this phase. Say: 'Before we discuss as a class, every group must write their answers in their journals. You have 8 minutes.' Set timer. 2. Circulate, reading written answers. Look for: use of the words 'magnetisation', 'demagnetisation', 'induced', 'current', 'coil', 'field strength'. If a group's answer to Q1 does not mention 'movement' or 'circuit breaking', prompt: 'What do both situations have in common about what has STOPPED?' WAIT 10 seconds. 3. After 8 minutes, cold-call one group for Q1. Ask the group's quietest member (identified during circulation): 'Read me your group's answer to Question 1.' WAIT 12 seconds. After the response, ask the class: 'Does anyone want to add to or challenge that answer?' Accept 2 responses. 4. Cold-call a DIFFERENT group for Q2. Ask: 'How many coil turns do you think the Kericho generator has compared to your iron nail?' Accept any reasonable estimate. Follow with: 'What sub-strand term describes the relationship between number of turns and the induced e.m.f.?' WAIT 12 seconds. Expect: 'factors affecting magnitude of induced e.m.f.' or 'number of turns of coil'. 5. Consolidate on the board: draw a two-column table headed 'Electric Bell' and 'Kericho Generator'. Ask the class to call out parallel features while you fill the table (iron core / copper coil / switching / movement essential /	Phenomenon-connected annotation with comparative analysis (NGSS Practices 6 — Constructing Explanations, and 8 — Obtaining, Evaluating and Communicating Information). The two-column comparison table is a 'bridging representation' — it makes the structural analogy between the miniature bell and the industrial generator visible and recordable. Writing answers before speaking prevents the most vocal learners from dominating the explanation phase, ensuring all groups engage with the reasoning. The vocabulary probe during circulation (checking for 'magnetisation', 'demagnetisation', 'induced') gives the teacher a real-time language audit.	Observable indicators: (1) Every group's annotated circuit diagram includes all five labelled elements (a–e listed above). (2) Written answers to Q1 include reference to movement/circuit-breaking being essential in both cases. (3) Written answers to Q2 include at least one factor affecting e.m.f. magnitude (number of turns, current strength, or strength of magnet). (4) The two-column comparison table is copied into every project journal before the end of the phase. Teacher collects 2 journals at random (from different ability groups) and reviews during Phase 4 to inform Phase 5 support.

		current produced). This takes 3 minutes. 6. Say: 'Copy this comparison table into your project journal. It will help you update your model in Phase 5.'		
Phase 4 — Driving Question Board (DQB) Update: What new evidence have we gathered? 📹 **VIDEO:** Magnetism, light and the magneto optic Kerr effect Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB (established in Lesson 1 and updated each lesson) is displayed at the front. Each group writes on a sticky note (or card) two things: (1) ONE piece of new evidence from today's bell project that helps answer the driving question — 'How does moving a wire through a magnetic field produce electricity, and how is this principle powering homes, farms and industries across Kenya?'; (2) ONE new question that today's project has raised for them. Groups post their notes on the DQB under the two columns: 'NEW EVIDENCE' and 'NEW QUESTIONS'. The class then does a 3-minute silent read of all new contributions. The teacher facilitates a 4-minute whole-class discussion to identify: (a) which evidence best connects to the phenomenon; (b) which new question is most important to pursue in Lesson 8 (the final evidence-gathering lesson). The DQB is photographed for the ARES platform record.	1. Distribute sticky notes/cards to each group. Say: 'You have 4 minutes. Write your evidence statement and your new question. Be specific — reference what you built today.' 2. While groups write, re-read the driving question aloud slowly: 'How does MOVING a wire through a magnetic field produce electricity — and how is this PRINCIPLE powering homes, farms and industries across Kenya? What does your bell tell us about that question?' 3. Groups post notes. Facilitate 3-minute silent read: 'Stand up, walk to the DQB, read everything posted. Do not talk.' 4. After the silent read, ask the class: 'Which piece of evidence posted today is most directly connected to the Kericho generator?' Cold-call 2 groups for responses. WAIT 12 seconds each. 5. Point to the 'NEW QUESTIONS' column: 'Look at the questions your classmates raised. Which one do you think we MUST answer in Lesson 8 to fully answer our driving question?' Take a show-of-hands vote across all questions posted. Circle the top vote-getter. Say: 'We will come back to this in Lesson 8.' 6. Read aloud the driving question one final time: 'By now, how confident are you that you can answer this question? Show me with fingers: 1 = not at all confident, 5 = very confident.' Count the distribution. Note aloud: 'Most of us are at [X]. Lesson 8 is your chance to reach 5.'	Public knowledge-building through the Driving Question Board (NGSS Practice 7 — Engaging in Argument from Evidence). The DQB is a cumulative sensemaking tool: each lesson's evidence is layered onto previous lessons', making the trajectory of understanding visible. The silent read norm prevents social pressure from favouring louder voices. The show-of-hands confidence check is a rapid metacognitive formative tool that makes self-assessment visible to both teacher and learners.	Observable indicators: (1) Every group posts exactly one evidence statement and one new question — teacher checks for blank or off-topic cards and prompts revision on the spot. (2) Evidence statements posted reference at least one sub-strand concept by name (e.g. 'magnetisation', 'make-and-break', 'electromagnet', 'demagnetisation'). (3) The confidence distribution from the finger vote: if more than 40% of learners show 1 or 2 fingers, the teacher notes the most commonly cited gap (from the 'NEW QUESTIONS' column) and adjusts the Lesson 8 opening activity to address it directly.

Phase 5 — Model Building: Revise the cumulative class model 📹 **VIDEO:** [Magnetism, light and the magneto optic Kerr effect] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Study of Space by the Electromagnetic Spectrum (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class maintains a cumulative wall model (started in Lesson 1) that represents the anchoring phenomenon — showing the Kericho student, the bicycle dynamo, the generator room, the national grid line, and the dormitory lights. In this phase (10 minutes), each group receives a model revision card and adds to the class model in one of four assigned roles: Group A labels the make-and-break cycle on the generator diagram; Group B draws an analogy arrow connecting the bell's iron nail electromagnet to the generator's iron core stator; Group C annotates the model to show that stopping movement = stopping current (connecting the dynamo-held-still observation to the bell-armature-held observation); Group D adds a new panel to the model showing a real Kenyan application introduced by today's lesson — specifically the electric bell used in Kenyan school bell systems (e.g. the end-of-lesson bell at their own school) — and labels which sub-strand concepts it demonstrates. After groups add to the model, the teacher facilitates a 4-minute whole-class model tour: learners walk past the updated model and each group explains their addition in exactly 30 seconds.	1. Assign group roles before distributing model revision cards. Say: 'Each group has a specific job on the class model. Your job is to ADD to what is already there — do not remove or cover previous contributions.' 2. Distribute model revision cards (pre-prepared with the specific task for each group). Allow 6 minutes for model additions. Circulate and check that Group C's annotation explicitly uses the word 'demagnetisation' — if not, prompt: 'What is the scientific term for what happens to the nail when the circuit breaks?' WAIT 10 seconds. 3. After 6 minutes, initiate model tour: 'Groups, step to the model. Group A — you have 30 seconds to explain your addition. Go.' Time each group strictly at 30 seconds. 4. After all four groups have presented, stand back and address the whole class: 'Look at our model from Lesson 1 to Lesson 7. What is the single most important idea that now connects the bicycle dynamo, the Kericho generator, and the electric bell you built today?' WAIT 15 seconds. Cold-call 2 individual learners (not group spokespersons). Accept and paraphrase: 'So the key idea is that [learner's words] — which in scientific language we call [electromagnetic induction / controlled magnetisation and demagnetisation].' 5. Close the lesson: 'In Lesson 8 — our final evidence-gathering lesson — we will look at how the applications of electromagnetic induction reach beyond bells and bicycle dynamos to power Kenya's homes, Kericho tea factories, Kisumu fish-processing plants and the Nairobi industrial area. Come ready to	Cumulative model revision with jigsaw presentation (NGSS Practices 2 — Developing and Using Models, and 4 — Analysing and Interpreting Data). Assigning different groups distinct model-building roles ensures that the cumulative model is genuinely collaborative and that every learner has a stake in a specific part of the explanation. The 30-second timed group presentation creates accountability and rehearses scientific communication. The teacher's final cold-call question ('what single idea connects all three devices?') targets the overarching explanatory principle — electromagnetic induction as the unifying concept — and ensures the lesson closes on the correct scientific idea rather than on procedural recall.	Observable indicators: (1) The class model now contains all four group additions, each referencing at least one labelled sub-strand concept. (2) Group C's annotation uses the word 'demagnetisation' correctly. (3) Group D's Kenyan application panel correctly identifies the electric bell as an application of controlled magnetisation/demagnetisation. (4) The two cold-called individual learners in the final question are able to articulate the connecting idea (movement → magnetisation/induction → current) in their own words without prompting. (5) Project journal homework reminder is noted; teacher collects journals in Lesson 8 for summative project assessment using the KICD assessment rubric (Exceeds / Meets / Approaches / Below Expectations) on the criterion: 'Ability to experiment on electromagnetic induction using the stated procedure.'

		connect everything.' 6. Remind learners to complete their project journal entries as homework if not finished: annotated diagram, Q1 and Q2 answers, and one reflection sentence: 'What sub-strand concept am I most confident about now, and which one do I still want to understand better?'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record responses in your lesson preparation notes:

1. CONCEPT ACHIEVEMENT: Did the majority of groups successfully build a working electric bell? If not, was the failure predominantly due to: (a) poor circuit connections (insulation not stripped); (b) incorrect gap between armature and nail; or (c) misunderstanding of the make-and-break mechanism? What does this tell you about which sub-strand concepts need reinforcement in Lesson 8?

2. PHENOMENON CONNECTION: During Phase 3, were learners able to independently articulate the connection between the bell's make-and-break cycle and the continuous rotation of the Kericho generator coils? Did the two-column comparison table help? If learners struggled with Q2 (the generator analogy), consider opening Lesson 8 with a 5-minute revisit of the factors affecting magnitude of induced e.m.f. before moving to applications.

3. EQUITY OF PARTICIPATION: During cold-calls, did the same learners respond each time? Note names of learners who have not yet been cold-called across Lessons 1–7 and prioritise them in Lesson 8. Did the group role assignments in Phase 5 ensure that learners who are typically quieter had a speaking role during the model tour?

4. LANGUAGE: Did learners use the correct KICD sub-strand vocabulary (magnetisation, demagnetisation, induced electromotive force, make-and-break, armature, electromagnet) in their written journal entries and oral contributions? If vocabulary was imprecise, prepare a vocabulary wall card for display in Lesson 8.

5. SAFETY: Were all safety protocols observed (low voltage only, no mains connection, wire handling)? Note any incidents or near-misses in the incident log.

6. DQB QUALITY: Were the evidence statements posted on the DQB specific and concept-linked, or vague ('we learned about magnets')? If vague, revise the DQB sticky-note prompt for Lesson 8 to require learners to complete the sentence stem: 'Today's evidence shows that [concept] because [observation from bell project]...'

7. PROJECT JOURNAL: Were annotated diagrams complete and accurate? Identify 2–3 common diagram errors (e.g. missing break-contact label, armature drawn as fixed) to address at the start of Lesson 8 using anonymous examples from collected journals.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Groups built a simple electric bell using an iron nail, copper wire, dry cell, thin metal strip (armature), wooden base and drawing pins. They observed that when the circuit was closed, the nail electromagnet attracted the metal strip; when the strip touched the nail, the circuit broke; the nail demagnetised; the strip sprang back; the circuit closed again — producing a repeated ringing sound. Groups that added more coil turns or increased current observed a louder, faster ring. Groups that held the armature against the nail observed that ringing stopped immediately — directly paralleling the Kericho student holding the bicycle dynamo still.
What did I learn?	The electric bell works because of controlled, repeated magnetisation and demagnetisation of soft iron. The iron nail becomes an electromagnet (magnetisation) when current flows through the coil; it attracts the armature and breaks the circuit; without current the nail loses its magnetism (demagnetisation) and the armature springs back to close the circuit again. This make-and-break cycle is the same principle that governs the switching of polarity in a generator's rotating coil. The number of coil turns and the current

	strength (factors affecting the magnitude of induced e.m.f.) directly affect how strongly and quickly the cycle operates — connecting today's bell directly to the Kericho generator, which uses thousands of coil turns and powerful electromagnets to produce enough current for the entire dormitory.
How does this explain the phenomenon?	Movement (or current flow) is essential to electromagnetic action: when it stops, so does the electromagnetic effect — whether in a bicycle dynamo LED, a school electric bell, or the Kericho generator. The bell's electromagnet switching on and off is a miniature, visible demonstration of the same controlled magnetisation and demagnetisation that occurs continuously in industrial generators. This principle — underpinning the KICD key inquiry question 'How is the study of electromagnetic induction important in our daily life?' — powers school bell systems, water pumps on Rift Valley farms, fish-processing equipment in Kisumu, and the national grid that supplies Kenya's homes and industries.

LESSON 8 (40 minutes): Model Building, DQB Closure and Final Assessment: Answering the Driving Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning from the sub-strand by constructing an annotated generator model, closing the Driving Question Board with evidence-based explanations, and demonstrating mastery of how electromagnetic induction powers homes, farms and industries across Kenya.
Knowledge	Learners can explain how a generator converts mechanical motion into electrical energy through electromagnetic induction; state all four factors that affect the magnitude of induced e.m.f. (speed, magnetic field strength, number of turns, angle of movement); and describe how these principles operate in real Kenyan energy systems including the Olkaria Geothermal Plant and the Lake Turkana Wind Farm.
Skills	Learners can construct and annotate a diagrammatic model of a generator showing coil, magnet, slip rings and external circuit; write a structured scientific explanation answering the driving question using evidence gathered across all eight lessons; and evaluate their own initial predictions against current evidence-based understanding.
Attitudes	Learners demonstrate pride in Kenyan innovation and energy solutions, intellectual honesty in comparing early predictions with current understanding, and collaborative responsibility in finalising the Driving Question Board as a shared class record of learning.
Key Inquiry Question	How is the study of electromagnetic induction important in our daily life?
Purpose in Storyline	This lesson is the culminating sensemaking event of the sub-strand storyline. The Kericho student puzzle introduced in Lesson 1 is now fully resolved: learners have moved from confusion about spinning copper coils to a complete, evidence-based model explaining how mechanical rotation induces electrical current. The DQB transitions from questions to answers, and learners articulate connections between the bicycle dynamo, the electric bell project, and Kenya's national grid.
Safety Notes	

B. LESSON OVERVIEW
Learners open by revisiting their very first written predictions about the Kericho generator from Lesson 1, then use a structured model-building task to produce an annotated generator diagram that synthesises all sub-strand concepts. The Driving Question Board is formally closed as groups replace prediction cards with evidence-based explanation cards using language developed across Lessons 1 through 7. Learners write a final structured explanation answering the driving question, complete a short practical assessment of the electromagnetic induction experiment procedure (demonstrating with galvanometer, conductor and U-shaped magnet), and close with a Kenya energy future reflection connecting Olkaria, Lake Turkana and local community electrification to the single principle discovered in Lesson 4: moving a conductor in a magnetic field induces an e.m.f.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Introduction to]()	Learners receive their original Lesson 1 prediction cards or notebooks and silently re-read	Distribute or direct learners to their Lesson 1 prediction notes. Say: 'Before we build our final model	Elicit-before-explain revisit: surfacing original misconceptions alongside current understanding	Teacher scans written before-and-after sentences during circulation. Look for: correct use of the phrase

experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	what they wrote about the source of electricity in the Kericho generator. Each learner then writes three sentences in their notebook: (1) what they predicted in Lesson 1, (2) what they now know to be correct, and (3) one specific piece of experimental evidence from any lesson that changed their thinking. A think-pair-share follows: learners compare their before-and-after understanding with a partner seated beside them, noting where their thinking has shifted most dramatically. Selected pairs share aloud. The teacher records two contrasting before-and-after statements on the board under the headings THEN and NOW as anchors for the entire lesson.	today, I want you to travel back in time to Lesson 1. Read what you wrote about why the Kericho student could see electricity coming from spinning coils. Do not change anything — just read.' Allow 2 minutes of silent reading. WAIT TIME: 2 minutes silent reading, no prompts. Then say: 'Now write three sentences: what you predicted then, what you know now, and one experiment that changed your mind. You have 3 minutes.' WAIT TIME: 3 minutes independent writing. Circulate and note interesting before-and-after pairs for sharing. After think-pair-share, select two or three pairs and ask: 'What was the biggest shift in your thinking — and what evidence caused that shift?' Record THEN and NOW statements verbatim on the board. Say: 'These two columns show the power of investigation. By the end of today, every question on our Driving Question Board will move from the THEN column to the NOW column.'	creates cognitive contrast that motivates model building. The THEN-NOW board display anchors subsequent phases to the storyline arc established in Lesson 1.	'induced e.m.f.' in the NOW sentence; reference to at least one specific experimental condition (speed, turns, field strength or angle); and connection to the Kericho phenomenon. Flag any learner still using battery or chemical-reaction language for targeted support during the Explain phase.
Observe Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12	The teacher performs a live 3-minute demonstration at the front of the class using the same apparatus from Lesson 4: U-shaped magnet, copper conductor connected to a sensitive galvanometer. The teacher systematically reproduces all four conditions tested in Lesson 4 — conductor stationary, moved parallel, moved vertically upward, moved at an angle — while also varying speed and using a coil of multiple turns (from Lesson 5 apparatus) to show the factor effect. Learners watch silently for 90 seconds, then record observations in a two-column	Set up apparatus before class begins. Begin by saying: 'I am going to run through our key experiment one final time. Watch everything — the galvanometer needle, the direction of my movement, the speed, the number of turns. Your job is to observe and record — not yet to explain.' Perform the demonstration deliberately and slowly, pausing between each condition. WAIT TIME: After each condition, pause 10 seconds in silence before moving to the next, giving learners time to write. After the demonstration say: 'You have 2 minutes to complete your	Teacher demonstration as anchor observation: replaying the core experiment before model building ensures all learners have a shared, accurate observational reference. The two-column table with factor labels is a structured note-making strategy that prepares learners to populate their annotated generator diagram in the next phase.	Collect one observation table from each group at the end of this phase (or teacher circulates and checks). Success criterion: all four conditions recorded, galvanometer behaviour described accurately (deflects left, deflects right, no deflection, large deflection, small deflection), and at least three factor labels correctly assigned. Address any confusion about direction of deflection immediately before the Explain phase begins.

🔍 Search ARES for similar readings	table: CONDITION and GALVANOMETER BEHAVIOUR. After recording, learners annotate each row with a label from the factor list (speed, field strength, number of turns, angle). This structured observation bridges the practicals from Lessons 4 and 5 into a single coherent picture ready for model building.	observation table and add the factor labels to each row.' WAIT TIME: 2 minutes. Then ask: 'Which condition produced the largest deflection? Raise your hand if you can say why in one sentence.' Cold-call two learners. If a learner gives a vague answer, say: 'Interesting — can you connect that to a specific variable we investigated in Lesson 5?' This redirects to evidence without dismissing the response.		
Explain Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually produce a final annotated generator diagram. Each learner draws a simple AC generator (rectangular coil, two poles of a U-shaped magnet, slip rings, brushes, external circuit with a lamp) and labels: (1) the rotating coil, (2) the magnetic field direction, (3) the direction of induced current, (4) all four factors affecting magnitude of induced e.m.f. with a brief note on how each factor increases the e.m.f., and (5) a Kenya-context annotation connecting the diagram to one real Kenyan energy system — Olkaria Geothermal Plant turbines, Lake Turkana Wind Farm blades, or the school generator in Kericho. Learners then write a structured final explanation of 6 to 8 sentences answering the driving question: 'How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya?' The explanation must use the terms: electromagnetic induction, induced e.m.f., magnetic field, conductor, factors, and at least one Kenyan energy context. Pairs then peer-review each other's diagram and explanation using a checklist of the	Distribute blank A4 paper or exercise book pages. Say: 'You are now going to draw the final model that explains everything the Kericho student wondered about in Lesson 1. Your diagram must show how the generator works and your written explanation must answer our driving question using evidence from our experiments.' Display the five labelling requirements and six required terms on the board or a chart. Say: 'You have 10 minutes. This is individual work — every learner produces their own diagram and explanation.' WAIT TIME: Allow 10 minutes of focused individual work. Circulate constantly, using targeted questions: 'Where in your diagram does the speed of rotation affect the e.m.f.?' or 'How does the Olkaria turbine connect to what you have drawn?' After 10 minutes, direct peer review: 'Exchange with your partner. Use the checklist on the board to give two specific pieces of feedback — one thing that is strong and one thing that could be improved.' WAIT TIME: 3 minutes peer review. Invite two or three volunteers to share their Kenya-context annotations with the class.	Model construction as consolidation: producing an annotated scientific diagram requires learners to organise, prioritise and communicate all sub-strand knowledge simultaneously. The structured written explanation ensures verbal and visual representations are aligned. Peer review with a checklist makes assessment criteria transparent and shifts evaluative responsibility to learners, building metacognitive skill.	Teacher collects all diagrams and written explanations at the end of this phase as the primary summative evidence for the sub-strand. Assess against: (a) accuracy of generator diagram components, (b) correct labelling of all four factors, (c) correct direction of induced current shown, (d) coherent 6-to-8-sentence explanation using all six required terms, and (e) meaningful Kenya energy context. Peer review checklists are also collected as evidence of collaborative critical thinking.

	five labelling requirements and the six key terms.	Say: 'Notice how the same principle — a conductor moving in a magnetic field — powers a bicycle dynamo in Kericho AND the turbines at Olkaria AND the windmills at Lake Turkana. One principle, three very different machines.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Study of Space by the Electromagnetic Spectrum (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The Driving Question Board, which has been built up since Lesson 1, is now formally closed. Groups are assigned a cluster of DQB question cards that were posted during Lessons 1 through 7. Each group reads their assigned questions aloud, then writes a concise evidence-based answer card for each question using findings from experiments conducted across the sub-strand. Answer cards are attached beside or over the original question cards on the board. The class then conducts a 3-minute gallery walk, reading the completed board. Learners place a green dot sticker beside the answer they find most satisfying and a yellow dot beside any answer they feel still needs more depth. The teacher facilitates a 2-minute whole-class debrief on the yellow-dotted cards, modelling how scientific understanding is always open to further investigation. The board is photographed as a permanent class record.	Direct learner attention to the DQB and say: 'This board has held our questions since Lesson 1. Today we close it — not by throwing the questions away, but by answering them with evidence. Every group has been assigned a cluster of questions. Your job is to write one answer card per question using specific evidence from our experiments.' Distribute sticky notes or card strips. Allow 4 minutes for group answer writing. WAIT TIME: 4 minutes group work, teacher circulates checking that answers cite specific experimental evidence rather than general statements. During the gallery walk, say: 'Walk silently, read every answer card, and place your green dot on the answer that best satisfies your curiosity. Place your yellow dot where you still have questions.' After the gallery walk, point to yellow-dotted cards and say: 'These yellow dots are not failures — they are the beginning of the next investigation. Scientists always find new questions inside every answer.' Ask: 'What new question has this sub-strand opened for you about Kenya's energy future?' WAIT TIME: 30 seconds thinking time before taking responses.	DQB closure as epistemic ritual: formally transitioning question cards to answer cards externalises the class's collective knowledge growth and validates every learner's contribution to the inquiry. Dot-sticker evaluation introduces evidence-quality judgement and models scientific humility. Photography of the completed board creates a sharable artefact for school display or parent communication.	Teacher evaluates answer cards for: specificity of experimental evidence cited, correct use of sub-strand vocabulary, and logical connection between the original question and the answer provided. Yellow-dotted cards reveal residual misconceptions or genuine curiosity that can inform future lesson planning or cross-strand connections (e.g., to physics electricity topics in later grades).
Model Building Phase	In this closing phase, the lesson returns explicitly to the anchoring phenomenon one final time. The	Re-read the Kericho phenomenon slowly and clearly. Say: 'This is where we started. Eight lessons	Return-to-phenomenon as epistemic closure: re-reading the original scenario and writing to an	Letters are collected as a supplementary assessment artefact. Assess for: accurate use

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D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did learners demonstrate a clear conceptual shift from Lesson 1 predictions to Lesson 8 explanations — were the THEN-NOW contrasts substantive or superficial? (2) Were annotated generator diagrams accurate in showing induced current direction and all four factors — if not, which factor was most commonly missing or mislabelled? (3) Did the DQB closure reveal any persistent misconceptions that were not resolved during the eight lessons — particularly around why stillness produces no current or why direction of movement matters? (4) Were Kenya energy contexts (Olkaria, Lake Turkana, school generator) genuinely integrated into explanations or added as superficial labels? (5) Did the letter-to-Kericho-student task reveal any learners who can explain the phenomenon verbally but struggle with written scientific communication — and how will this inform support in the next sub-strand? (6) Consider whether the electric bell project from Lesson 7 was effectively connected to the generator model in this lesson — the make-and-break magnetisation cycle should have appeared in at least some letters or diagrams. (7) Note which learners demonstrated leadership during DQB closure and peer review for recognition in co-curricular science activities or science club.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that moving a copper conductor in a magnetic field causes a galvanometer to deflect, and that the deflection is larger when the conductor moves faster, the magnet is stronger, more coil turns are used, and the conductor moves perpendicular to the field. We also observed that stillness produces zero deflection — the conductor must be moving relative to the magnetic field for any current to be induced.
What did I learn?	We learned that electromagnetic induction is the process by which mechanical motion produces electrical energy without any battery or chemical reaction. We learned that the magnitude of the induced e.m.f. depends on four factors: speed of movement, strength of the magnetic field, number of turns of wire, and angle of the conductor relative to the field. We also learned that this single principle explains how the Kericho school generator, the bicycle dynamo, the Olkaria Geothermal Plant turbines and the Lake Turkana Wind Farm all produce electricity for Kenyan homes, farms and industries.
How does this explain the phenomenon?	We can now fully explain why the Kericho student sees electricity produced from spinning coils with no battery: the mechanical rotation of the coil in the magnetic field continuously changes the flux through the coil, inducing an e.m.f. that drives current through the external circuit. The faster the coil spins, the greater the induced e.m.f. and the brighter the lights. The same principle explains why the bicycle dynamo LED goes out when the wheel stops — without relative motion between the conductor and the magnetic field, no e.m.f. is induced and no current flows. Our electric bell project from Lesson 7 showed that controlled magnetisation and demagnetisation of soft iron is the same switching mechanism used in generator coils, confirming that all sub-strand concepts are interconnected aspects of one fundamental electromagnetic phenomenon.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.